Which Parts of a Tool's Documentation Does a Model Actually Need?

A controlled ablation of tool schemas on BFCL and ToolSandbox multi-turn tasks

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Abstract

Function-calling interfaces document every tool with a name, a description, parameter names and parameter descriptions. We ask which of these a model actually depends on. Holding the Berkeley Function Calling Leaderboard (BFCL) multi-turn base tasks, backends and scorer fixed, we ablate one component at a time, and then all four, for four models (GPT-6 Luna, Claude Haiku 5.5, Qwen3-30B-A3B and Gemma 4 26B-A4B) on the same 200 tasks, with paired statistics and a repeated baseline to measure run-to-run noise. Parameter descriptions are the only component whose removal significantly hurts every model: task success falls by 14 to 33 percentage points, while removing tool names costs nothing and removing tool descriptions costs 2.5 to 6 points, consistent in sign across models and runs but not separable from run-to-run noise with one run per condition. Trajectory analysis locates the damage at argument construction rather than tool selection and traces it to a handful of parameters whose descriptions carried an allowed-value list, a value format or an entity type that the schema did not encode; the six most frequent account for the drop at task-level resolution, with a small tail from other prose-only enums. Moving those constraints into `enum`, `pattern` and better parameter names, with every description left blank, restores baseline success for all four models. With all documentation removed, success is at most 1% under BFCL's evaluator and no model recovers a task after an execution error. The mechanism predicts what happens on a second benchmark: on Apple's ToolSandbox, whose docstring-generated descriptions carry no constraint that the schema lacks, removing parameter descriptions costs no model anything, while removing everything costs 39 to 49 points of similarity. Switching reasoning off for the two models that reason by default, and on for one that does not, leaves every drop in place. On this benchmark, parameter descriptions matter not as prose but as the place where structural constraints happened to live.

1 Introduction

Tool use is now the default way to connect a language model to the outside world. Whether through provider-native function calling (OpenAI, 2023), the Model Context Protocol (Anthropic, 2024), or bespoke agent frameworks, the model is shown a list of tools expressed as JSON Schema (Wright et al., 2022): a name and a natural-language description for each tool, and for each parameter a name, a type, optional constraints and another natural-language description. Practitioners spend considerable effort writing these descriptions, and the literature contains advice on how to write them (Hsieh et al., 2023; Qu et al., 2025a; Yuan et al., 2025). Yet there is little controlled evidence on which component of a tool's documentation a model actually relies on when it works through a realistic, stateful, multi-turn task.

This paper reports such an experiment. We take the 200 *multi-turn base* tasks of the Berkeley Function Calling Leaderboard (BFCL) (Mao et al., 2024; Patil et al., 2025), keep every part of the benchmark fixed except the documentation shown to the model, and ablate that documentation

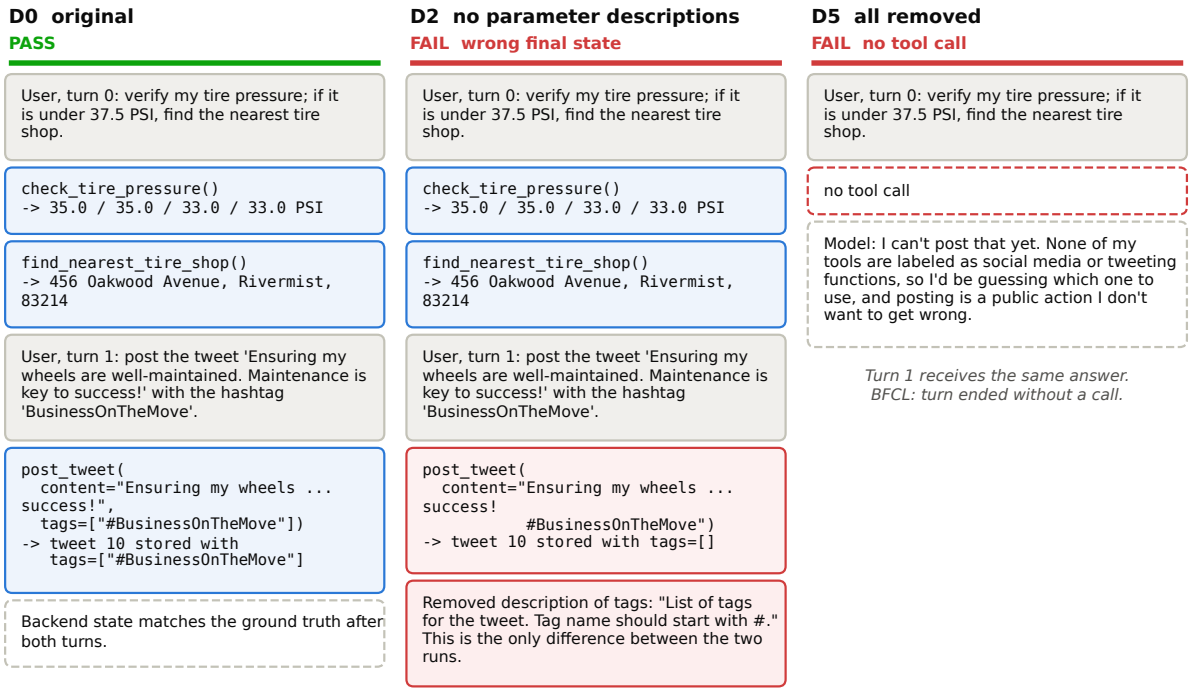


Figure 1: One task, three documentation conditions. Claude Haiku 5.5 on BFCL task `multi_turn_base_78`. With the original schema (D0) the model checks the tires, finds the shop and posts the tweet with the hashtag in the `tags` list. With parameter descriptions removed (D2) the first turn is identical, but the hashtag is folded into the tweet body, the backend stores no tags, and the final state no longer matches the ground truth. With names and descriptions removed (D5) the model declines to guess which tool posts and never acts. Section 4.5 and Appendix B give the full trajectories.

one component at a time. Six conditions (Table 1) remove tool descriptions (D1), parameter descriptions (D2), tool names (D3) or parameter names (D4) individually, or all four together (D5), against the original schema (D0); two follow-up conditions (D6, D7) then test what the removed parameter descriptions were contributing. Types, required lists, enums, defaults and the response schema are preserved in every condition, as are tool outputs and backend state, so the experiment measures the contribution of documentation rather than the information an agent can obtain through interaction. Four models spanning two frontier small models and two open-weight mixture-of-experts models ran every task under every condition, and because the same tasks appear in every condition the comparisons are paired at the task level. The six single-component conditions were then repeated on Apple's ToolSandbox (Lu et al., 2024), which differs from BFCL in its backends, its scorer and in simulating the user with an LLM, and the main comparisons were repeated with each model's reasoning switched explicitly on or off.

Beyond the headline success rates, we recorded every raw request and response, every tool call, every execution result and the backend state after every turn, so that we can ask *where* in the tool-use process documentation matters: selecting the right tool, constructing the right arguments, or recovering from errors. Our findings are:

- **Parameter descriptions are the component the models depend on** (Section 3). Removing them (D2) costs 14.0 to 33.0 percentage points for every model, significant after Bonferroni correction. Replacing tool names with `tool_NN` (D3) stays within the run-to-run noise measured by a repeated D0 run. Removing tool descriptions and response-schema prose (D1) costs 2.5 to 6.0 points, in the same direction for every model; no D1 difference survives correction, so we report it as a possible small cost rather than a null result. Replacing parameter names with `arg_N` (D4) is harmless for GPT-6 Luna and Gemma 4, costs Qwen3 11.5 points, and produces an apparent gain for Claude Haiku that falls within

the noise band of the repeated baseline.

- **The damage is at argument construction, not tool selection** (Section 4.1). In D2 the right tool is chosen for the first call as often as in D0, but the right tool *with* the right arguments falls by 11 to 15 points, and failures begin with a wrong-argument call rather than a missing tool.
- **The effect sits on a handful of parameters** (Section 4.2). Six parameters account for most of the new argument errors in every model; their descriptions carried an allowed-value list written only in prose (`order_type`: "Buy/Sell"), a format convention (`tags` start with #) or an entity type the name hides (`cityA` is a zip code). The D2 drop on the 91 tasks that use them is 36 to 70 points; on the other 109 tasks it is within ±5 points. A tail of other prose-only enums (`lockDoors.door`, `activateParkingBrake.mode`) adds a few points.
- **Structure substitutes for the prose** (Section 4.6). Keeping every parameter description blank but moving the prose-only enums into `enum`, adding `pattern` for tags and mentions, and renaming the two zip-code parameters (D6) brings all four models back to baseline. Keeping only the six descriptions (D7) recovers most of the loss.
- **The mechanism predicts a second benchmark** (Section 3.5). On ToolSandbox's 91 base scenarios, none of whose 72 parameter descriptions carries a constraint the schema lacks, D2 changes mean similarity by between −6.3 and +2.0 points for the four models (none significant) and D5 costs 39 to 49 points, with the same split between models that fall silent and models that probe to the limit.
- **Reasoning does not change the effect** (Section 3.6). With the same weights and reasoning switched off (Luna, Haiku) or on (Gemma 4), the D2 drop stays within 5.5 points of the default run for every pair and remains significant, and Haiku's apparent D4 gain vanishes with reasoning off. Models reason two to eight times longer when documentation is removed and recover none of the loss.
- **Nobody compensates through exploration** (Section 4.3). With everything removed (D5) success is at most 1% under BFCL's 20-step, final-state evaluator; two models stop calling tools, one probes blindly to the step limit, and none recovers a task after an execution error.

Section 2 describes the benchmark, conditions, harness and metrics. Section 3 reports the main results and paired statistics, Section 4 analyses the trajectories to locate where documentation matters, and Section 5 interprets the findings. Section 6 places the study in the tool-learning literature, and Sections 7 and 8 close with limitations and conclusions.

2 Experimental setup

2.1 Benchmark

BFCL's `multi_turn_base` category (Mao et al., 2024) contains 200 hand-curated tasks. Each task is a conversation of one to seven user turns (median four) over a small set of stateful Python backends: a file system, a vehicle-control API, a trading bot, a travel API, a messaging API, a Twitter-like API, a ticketing API and a math API. A task involves one or two backends, and the model is shown every function of the involved backends, 18 to 39 tools per task (median 28). The ground truth lists, for each turn, the minimal set of calls that bring the backend into the required state (2 to 10 calls per task, mean 5.7). A turn is scored by BFCL's own multi-turn checker: the backend state after the turn must match the state produced by the ground-truth calls (ignoring private attributes), and for read-only requests the model's execution path must contain the

Table 1: Documentation conditions. "Generic" tool names are `tool_01`...`tool_NN` in BFCL's listing order; generic parameter names are `arg_1`...`arg_n` per tool, applied recursively to object-typed parameters. Parameter types, required lists, enums, defaults and the response-schema structure are kept in every condition. †D1 also removes the natural-language descriptions inside the response (return-value) schema, so it is an ablation of all tool-level prose, not of the one-line tool description alone. ‡D6 renames `cityA`/`cityB` to `zipcodeA`/`zipcodeB`; all other names are original. D6 and D7 were run after the main study, in response to its results.

	Tool names	Tool descriptions	Parameter names	Parameter descriptions
D0	original	original	original	original
D1	original	**removed**†	original	original
D2	original	original	original	**removed**
D3	**generic**	original	original	original
D4	original	original	**generic**	original
D5	**generic**	**removed**	**generic**	**removed**
D6	original	original	original‡	**removed**; constraints in schema
D7	original	original	original	**removed**; six kept (Table 8)

ground-truth outputs; an entry passes only if every turn passes (Mao et al., 2024). Within a turn the model may make up to 20 steps (model requests); exceeding the limit terminates the entry as a failure. We used BFCL at commit `6ea57973` (Gorilla LLM contributors, 2026) with its default temperature of 0.001, and did not modify the harness, the backends or the evaluator.

Second benchmark: ToolSandbox. To test whether the findings are specific to BFCL's backends and scorer, the single-component conditions D0 to D5 were also run on ToolSandbox (Lu et al., 2024), used unmodified at commit `c8571d7`. ToolSandbox differs from BFCL in three ways that matter here: the user is simulated on-policy by an LLM (GPT-4o-2024-05-13, as in the original paper, reached through OpenRouter like the agents), so the agent can ask for clarification and the conversation is not scripted; the tools are 34 Python functions over a phone-like world state (contacts, messages, reminders, settings, time and unit utilities) whose schemas are generated from docstrings, so every parameter description is a short Google-style "Args" line rather than BFCL's long prose; and the score is a similarity in $[0, 1]$, the mean over the scenario's milestones of how closely the best-matching point in the trajectory meets each milestone (exact match on identifiers, ROUGE-L on free text), rather than a binary verdict. ToolSandbox's 1 032 scenarios are 129 base scenarios each in eight variants that vary distraction tools and apply the benchmark's own tool augmentations; we take the 91 base scenarios that need none of the five tools backed by a paid web API, in the variant that offers the agent every tool that the scenario does not itself withhold (24 to 28 tools per scenario after removing those five; 28 in 73 of the 91), so that tool selection is part of every task as it is in BFCL. Our conditions are applied to the generated OpenAI-format schemas exactly as for BFCL, and the model's calls are translated back before the sandbox executes them. ToolSandbox executes each call as a Python statement named after the provider's tool-call id; two providers return ids that are not valid Python identifiers (a leading digit, or hyphens), which the unmodified harness turns into a syntax error on every call, so the harness rewrites such ids before execution and keeps them consistent in the history. When a model emits arguments that are not valid JSON, ToolSandbox aborts the scenario; like its own command-line runner we score such a scenario 0 (this happened in 27 of 2 184 plays, all Qwen3's: 26 with invalid JSON arguments, one with a tool name that was not in the list). We report the mean similarity (the benchmark's own metric) and the share of scenarios at similarity 1.0, and test paired differences with a sign-flip permutation test on the similarity and McNemar's test on the binary outcome.

2.2 Documentation conditions

Table 1 defines the six conditions. Removing a tool description also removes the natural-language descriptions inside BFCL's response (return-value) schema, since those describe what the tool does; the response schema's structure and types are kept. Parameter descriptions are blanked rather than deleted because BFCL's schema converter expects the key to exist. In conditions with generic names, the model's calls are translated back to the original tool and parameter names before BFCL executes and scores them, so the backend and the evaluator see exactly what they would see in D0; the chat history the model sees keeps the names it was given. Names the model invents are passed through untouched and fail inside BFCL's executor as they would in the unmodified benchmark. In D0 the tools sent to the model are byte-identical to a stock BFCL run against an OpenAI-compatible endpoint.

D1 to D4 isolate the contribution of one component each. D5 is a test of *reduced* interface semantics, not of zero semantics: the user's request, the tool outputs, the parameter types, requiredness and enum values all remain, and tools remain distinguishable by their argument structure.

D6 and D7 are follow-ups designed after the main analysis identified six parameters as the carriers of the D2 effect (Section 4.2). Both blank every parameter description exactly as D2 does. D6 then puts back, in structural form only, the information that BFCL's authors had written in prose: every parameter whose description contains an allowed-value list ("[Enum]: ..." or "(Buy/Sell)") receives a JSON Schema `enum` with those values (11 parameters across the vehicle, trading and travel backends), `post_tweet.tags` and `mentions` receive an item `pattern` of `^#` and `^@`, and `estimate_distance.cityA`/`cityB` are renamed `zipcodeA`/`zipcodeB`. D7 instead keeps the original descriptions of the six parameters of Table 8 and blanks all others. D6 measures whether structure can replace the prose; D7 measures how much of the D2 drop those six descriptions alone account for.

2.3 Models and harness

We evaluated four models through OpenRouter (OpenRouter, 2026), chosen to span two frontier small models and two open-weight mixture-of-experts models of similar size: GPT-6 Luna (OpenAI, 2026), Claude Haiku 5.5 (Anthropic, 2026), Qwen3-30B-A3B-Instruct-2507 (Yang et al., 2025) and Gemma 4 26B-A4B (Google DeepMind, 2026). All four were used in native function-calling mode. Provider routing was left to OpenRouter. Every Luna request was served by Azure and every Haiku request by Google; Gemma 4 was served by DeepInfra except for 7 of 1 200 trajectories with some requests routed to Makora; Qwen3 was served by SiliconFlow in every trajectory, but 336 of its 1 200 trajectories had at least one request routed to Nebius. Qwen3's success is lower on those mixed-provider trajectories in every condition (for example 47% against 57% in D0 and 17% against 34% in D2), so the Qwen3 numbers should be read as those of a two-provider deployment; the D2 effect is present on both subsets. No reasoning-effort parameter was set: 99% of Luna and Haiku trajectories contain reasoning tokens (674 and 664 per trajectory on average), while Qwen3 emitted none and Gemma 4 did so in 7 of its 1 200 trajectories (at most 280 tokens). Any contrast between the frontier and the open-weight pair is therefore confounded with reasoning; the within-model comparisons that carry the paper's claims are not. To test whether reasoning changes the documentation effect itself, three reasoning-controlled pairs were run after the main study (Section 3.6): the same model id with OpenRouter's `reasoning` parameter set explicitly, off for Luna and Haiku (which reason by default) and on for Gemma 4 (which does not), on all 200 tasks under D0 to D5. Qwen3-30B-A3B-Instruct-2507 has no reasoning switch (its thinking variant is a different checkpoint), so it has no pair.

The harness subclasses BFCL's OpenAI-compatible handler. It differs from a stock BFCL run only in routing requests to OpenRouter, applying the condition to the tool schemas, translating generic names back before execution, and recording one JSON file per trajectory containing the

task, the exact tools sent, the name maps, BFCL's result and inference log (with tool outputs and backend state after every turn), every raw HTTP request and response, and the evaluator's verdict. The main study (4 models × 6 conditions × 200 tasks = 4 800 trajectories) ran in about two hours of wall-clock time on 2 November 2026 with eight concurrent tasks per model-condition pair and four pairs in parallel; the two follow-up conditions (1 600 trajectories) ran the same afternoon and a repeated D0 run of Claude Haiku 5.5 (200 trajectories) about eight hours after the first, both with the same settings. The reasoning-controlled pairs ($3 \times 6 \times 200 = 3\,600$ trajectories) and the ToolSandbox runs ($4 \times 6 \times 91 = 2\,184$ plays, sixteen in parallel) ran on 3 November 2026. Four Qwen3 trajectories (one each in D0 and D3, two in D5) ended in a provider error and are counted as failures; dropping the three affected tasks pairwise instead changes no paired difference by more than 0.8 pp and no significance verdict.

2.4 Metrics and statistics

Task success is BFCL's own verdict and is the primary metric. All behavioural metrics are computed from the trajectory files and the ground truth without any model-based judging, using the following definitions. A *call* is a tool call BFCL decoded from the model's response; malformed calls end the turn and are not counted. A *step* is one model request; a plain-text reply also counts as a step and ends the turn. A call is *on-path* if its tool appears anywhere in the task's ground truth, *off-path* if it does not but executed without error, and *failed* if the backend returned an execution error. A *repeated* call has identical tool and arguments to an earlier call in the same task.

For each ground-truth call in turn t, a *tool hit* means the model called that tool in any step of turns 0 to t (matching is cumulative across the task, not per turn), and an *exact hit* means one of those calls had identical arguments. Two argument values match if they are equal, if both are numeric and differ by less than 10^{-9}, or if they are equal as whitespace-stripped strings; there is no case folding, so `"buy"` does not match `"Buy"`. One model call can satisfy several identical ground-truth calls, and each ground-truth call is credited at most once. *Tool-selection accuracy* is tool hits over all ground-truth calls; *argument accuracy given the right tool* is exact hits over tool hits. Because an exact hit requires every argument value to match, a semantically equivalent but differently spelled value counts as a miss; this makes argument accuracy conservative in every condition equally. For each failed task the *first divergence* is the earliest ground-truth call, in turn order, without an exact hit, classified as *wrong arguments* if the tool was called with other arguments or *tool never called* otherwise. When a ground-truth call was matched by the right tool but wrong arguments, the closest same-tool call is found and the parameters whose values differ are recorded; counting these in D2 and subtracting the D0 count isolates the errors that appear when parameter descriptions are removed. *Recovery rate* is the share of tasks containing at least one failed call that nevertheless passed.

Because every task appears in every condition, each ablation is compared with D0 by a paired analysis (McNemar, 1947): we report the two discordant cells (tasks passing only in D0 and tasks passing only in the ablation), the exact two-sided McNemar p-value computed from them, and a 95% confidence interval for the paired success-rate difference from a task-level bootstrap with 2 000 resamples (Efron, 1979). With 20 pre-planned comparisons (five ablations × four models) we use a Bonferroni threshold of $p < 0.0025$ (Dunn, 1961) and treat everything else as exploratory; the follow-up comparisons in Section 4.6 are reported with the same test but were not part of the pre-planned family.

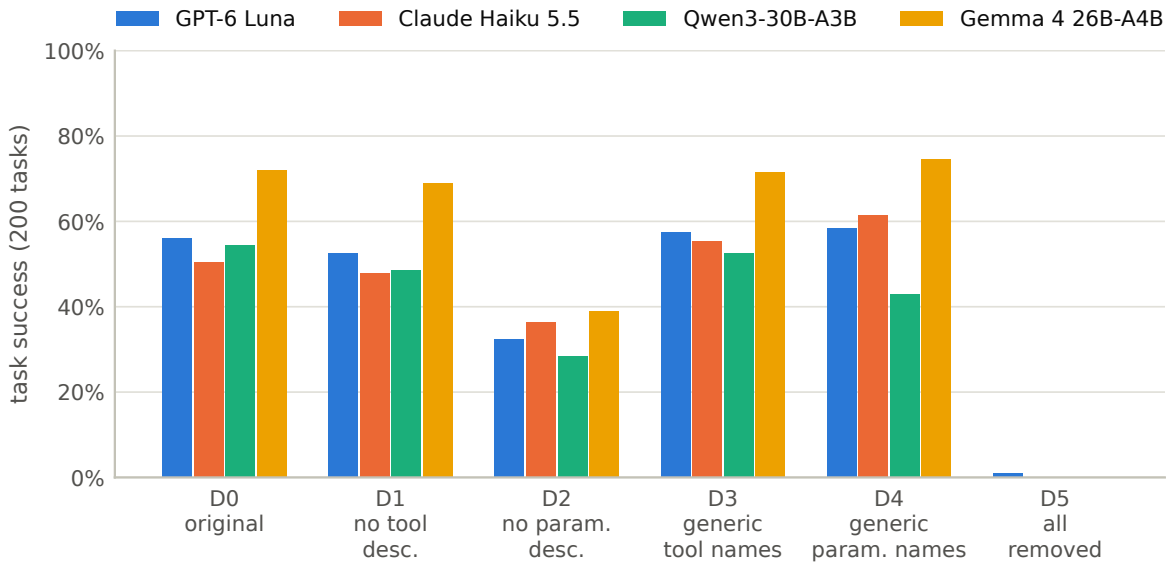


Figure 2: Task success rate by documentation condition. Each bar is 200 tasks; the same tasks appear in every bar, so bars within a model are paired.

Table 2: Task success (%) per condition, 200 tasks per cell.

Model	D0	D1	D2	D3	D4	D5
GPT-6 Luna	56.0	52.5	32.5	57.5	58.5	1.0
Claude Haiku 5.5	50.5	48.0	36.5	55.5	61.5	0.0
Qwen3-30B-A3B	54.5	48.5	28.5	52.5	43.0	0.0
Gemma 4 26B-A4B	72.0	69.0	39.0	71.5	74.5	0.0

3 Results

3.1 Documentation sensitivity

Figure 2 and Table 2 give the headline result. Every model shows the same shape: a modest dip for D1, a large drop for D2, no change for D3, and a collapse for D5. Gemma 4 has the highest baseline (72.0%) and the largest absolute D2 drop (33.0 pp). The heatmap in Figure 3 shows the same numbers as paired changes; a single-number sensitivity score per model is given in Appendix A (Table 11) but, because D2 dominates every model's sensitivity, it conceals more than it shows.

3.2 Paired task-level statistics

Table 3 shows that D2 and D5 are significant for all four models at the corrected threshold. The D2 discordant cells are lopsided: tasks that pass only in D0 outnumber tasks that pass only in D2 by 53 to 6 (Luna), 42 to 14 (Haiku), 63 to 11 (Qwen3) and 71 to 5 (Gemma 4). D1 and D3 have discordant cells of similar size in both directions for every model, which is what run-to-run noise at near-zero temperature looks like; their confidence intervals all include zero, although the four D1 differences all point the same way (-2.5 to -6.0 pp; see Section 3.3). The two D4 results that survive correction point in opposite directions: Haiku gains (29 tasks pass only in D4 against 7 only in D0; $p = 0.0003$) and Qwen3 loses (12 against 35; $p = 0.0011$). Haiku's 29 gained tasks are not concentrated anywhere: they span 16 combinations of backends, 19 of them involve the six parameters of Table 8 and 10 do not, and their D0 failures begin with a wrong argument in 12 cases, a never-called tool in 7 and no identifiable divergence in 10. Section 3.3 puts this result against a direct measurement of run-to-run noise.

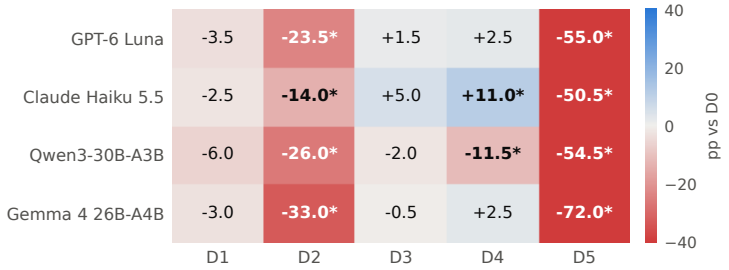


Figure 3: Change in task success relative to D0, in percentage points. Starred, bold cells are significant at the Bonferroni-corrected threshold $p < 0.0025$ (exact McNemar test, Table 3). Colour saturates at ± 40 pp.

Table 3: Each ablation versus D0 on the same 200 tasks. The two discordant cells carry the evidence; Δ is the paired success-rate difference in percentage points with a 95% task-level bootstrap interval (2 000 resamples); p is the exact two-sided McNemar test. The last column applies the Bonferroni threshold for 20 comparisons.

Model	Cond.	both pass	D0 only	abl. only	both fail	Δ pp	95% CI	McNemar p	$p < 0.0025$
GPT-6 Luna	D1	88	24	17	71	-3.5	[-9.5, 2.5]	0.3489	no
	D2	59	53	6	82	-23.5	[-30.0, -17.0]	<0.0001	**yes**
	D3	97	15	18	70	+1.5	[-4.0, 7.5]	0.7283	no
	D4	98	14	19	69	+2.5	[-3.0, 8.0]	0.4869	no
	D5	2	110	0	88	-55.0	[-61.5, -48.0]	<0.0001	**yes**
Claude Haiku 5.5	D1	79	22	17	82	-2.5	[-8.5, 3.5]	0.5224	no
	D2	59	42	14	85	-14.0	[-20.5, -7.0]	0.0002	**yes**
	D3	84	17	27	72	+5.0	[-1.0, 11.5]	0.1742	no
	D4	94	7	29	70	+11.0	[5.5, 17.0]	0.0003	**yes**
	D5	0	101	0	99	-50.5	[-57.5, -44.0]	<0.0001	**yes**
Qwen3-30B-A3B	D1	78	31	19	72	-6.0	[-13.0, 0.5]	0.1189	no
	D2	46	63	11	80	-26.0	[-34.0, -18.5]	<0.0001	**yes**
	D3	85	24	20	71	-2.0	[-8.5, 4.5]	0.6516	no
	D4	74	35	12	79	-11.5	[-18.5, -5.0]	0.0011	**yes**
	D5	0	109	0	91	-54.5	[-61.5, -48.0]	<0.0001	**yes**
Gemma 4 26B-A4B	D1	130	14	8	48	-3.0	[-7.5, 1.5]	0.2863	no
	D2	73	71	5	51	-33.0	[-40.5, -26.0]	<0.0001	**yes**
	D3	136	8	7	49	-0.5	[-4.5, 3.5]	1.0000	no
	D4	141	3	8	48	+2.5	[-0.5, 5.5]	0.2266	no
	D5	0	144	0	56	-72.0	[-78.5, -66.0]	<0.0001	**yes**

3.3 Run-to-run noise

The D1 and D3 discordant cells suggested that two runs of the same condition differ on roughly 15 to 30 tasks in each direction. Table 4 measures this directly: Claude Haiku 5.5's second D0 run passed 56.5% of tasks against 50.5% in the first, with 9 tasks passing only in the first run and 21 only in the second ($\Delta = +6.0$ pp, $p = 0.0428$). At temperature 0.001 the models are not deterministic, the backends are, so every discordance here is the model's. The gap between the two identical runs is comparable to the D1 and D3 gaps against either of them, and is itself nominally significant at $p < 0.05$ (though not at the corrected threshold). Four consequences follow. The D3 and (for three models) D4 differences are of the same size as this floor and should be read as null results. D1 is less clear: against the second D0 run Haiku's D1 gap widens to -8.5 pp, with a confidence interval that excludes zero ($p = 0.008$, above the corrected threshold), and D1 is negative for all four models against the first run. We read D1 as a possible cost of a few points that a single run per condition cannot resolve, not as a null result; the three reasoning-controlled reruns of Section 3.6 are negative as well. The D2 and D5 differences are an order of magnitude larger than it and remain significant against the second D0 run as well as the first (Haiku's D2 drop is 14.0 pp against the first run and 20.0 pp against the second).

Table 4: A repeated D0 run of Claude Haiku 5.5 against the first, and each ablation against both D0 runs. Same 200 tasks, same settings, run eight hours apart. The discordance between two identical runs is the floor against which single-condition differences must be read.

Comparison	both pass	first only	second only	both fail	Δ pp	95% CI	McNemar p
D0 run 1 vs D0 run 2	92	9	21	78	+6.0	[1.0, 11.5]	0.0428
D0 run 1 vs D1	79	22	17	82	-2.5	[-8.5, 3.5]	0.5224
D0 run 2 vs D1	86	27	10	77	-8.5	[-14.5, -3.0]	0.0076
D0 run 1 vs D2	59	42	14	85	-14.0	[-20.5, -7.0]	0.0002
D0 run 2 vs D2	61	52	12	75	-20.0	[-27.0, -13.0]	<0.0001
D0 run 1 vs D3	84	17	27	72	+5.0	[-1.0, 11.5]	0.1742
D0 run 2 vs D3	97	16	14	73	-1.0	[-6.0, 4.5]	0.8555
D0 run 1 vs D4	94	7	29	70	+11.0	[5.5, 17.0]	0.0003
D0 run 2 vs D4	100	13	23	64	+5.0	[-1.0, 11.0]	0.1325
D0 run 1 vs D5	0	101	0	99	-50.5	[-57.5, -44.0]	<0.0001
D0 run 2 vs D5	0	113	0	87	-56.5	[-63.0, -49.5]	<0.0001
D0 run 1 vs D6	87	14	21	78	+3.5	[-2.0, 9.5]	0.3105
D0 run 2 vs D6	95	18	13	74	-2.5	[-8.0, 3.0]	0.4731
D0 run 1 vs D7	85	16	25	74	+4.5	[-2.0, 11.0]	0.2110
D0 run 2 vs D7	96	17	14	73	-1.5	[-7.0, 3.5]	0.7201

Table 5: ToolSandbox results. Left: mean similarity (×100) over the 91 scenarios; right: share of scenarios solved exactly (similarity 1.0). Paired tests for every ablation against D0 are in Table 12, a per-category breakdown in Table 13.

	mean similarity						solved exactly (%)					
Model	D0	D1	D2	D3	D4	D5	D0	D1	D2	D3	D4	D5
GPT-6 Luna	81.5	82.9	81.2	85.4	82.5	32.3	25.3	31.9	25.3	31.9	29.7	14.3
Claude Haiku 5.5	80.1	82.1	82.2	79.3	81.8	36.3	26.4	29.7	25.3	25.3	29.7	17.6
Qwen3-30B-A3B	65.0	60.8	58.7	62.4	69.0	26.3	8.8	14.3	9.9	11.0	14.3	14.3
Gemma 4 26B-A4B	75.5	73.7	76.5	69.5	75.0	27.3	19.8	13.2	19.8	12.1	17.6	16.5

Haiku's D4 gain, +11.0 pp and significant against the first D0 run, shrinks to +5.0 pp against the second ($p = 0.13$), with the two D0 runs themselves 6 points apart. We report it as suggestive at most and offer no mechanism for it; with Haiku's reasoning switched off it disappears entirely (Section 3.6).

3.4 Memorisation check

BFCL's multi-turn tasks predate all four models, so a model could in principle recognise the backends from their argument structure and call the original tool names even when shown `tool_NN`. Invented names pass through the harness untouched and would appear in the raw responses. Across the 1 515 to 3 927 tool calls each model made in D3 and D5, no model emitted a single original tool name, and across D4 and D5 no model emitted a single original parameter name; the only non-generic strings were eight malformed generic names from Qwen3 (`tool_6` for `tool_06`) and ten non-generic argument keys from Luna in D4 and D5. The D3 and D5 results therefore measure what the models do with the interface they were shown, not what they remember.

3.5 A second benchmark: ToolSandbox

ToolSandbox (Section 2) gives a different answer to the single-component questions and the same answer to the joint one (Figure 4, Table 5). Removing parameter descriptions (D2) changes mean similarity by −0.3, +2.0, −6.3 and +0.9 points for Luna, Haiku, Qwen3 and Gemma 4 (sign-flip $p \geq 0.07$; Table 12), and the share of scenarios solved exactly is unchanged for every

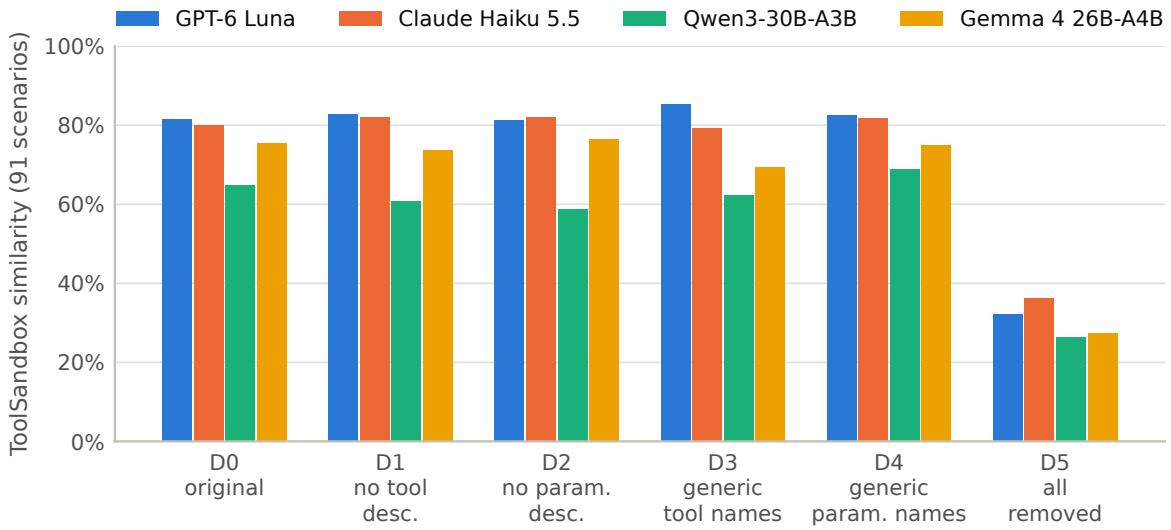


Figure 4: ToolSandbox mean similarity by condition (91 scenarios per bar, same four models). Removing any single component leaves every model within a few points of its baseline; removing all four (D5) costs 39 to 49 points.

model (discordant cells of 5/5, 2/1, 3/4 and 2/2). D1 (of mixed sign here: +1.4, +2.0, −4.2 and −1.8) and D4 are likewise within a few points, and so is D3 except for Gemma 4, which loses 6.0 points of similarity under generic tool names ($p = 0.003$) and seven exactly-solved scenarios against none gained ($p = 0.016$); this is the one single-component effect on ToolSandbox that reaches nominal significance, and it does not reach the corrected threshold. Removing everything (D5) costs 39 to 49 points of similarity for every model ($p < 0.0001$) and is, as on BFCL, an order of magnitude larger than anything a single component does.

The D2 null is what the BFCL analysis of Section 4.2 predicts rather than a contradiction of it. ToolSandbox's parameter descriptions are generated from docstrings and are short. Across the 72 parameters of the 28 tools an agent can see, none carries an allowed-value list that the schema does not: the only constraints written in prose are formats ("POSIX timestamp", "in 24hr format", "for example celsius"), and most of them restate what the parameter name already says (`reminder_timestamp_lowerbound`, `hour`, `from_unit`). The identifiers that the tools exchange (`person_id`, `reminder_id`) are returned by earlier calls rather than written by the user, so the agent copies them rather than constructs them. There is nothing in these descriptions for D2 to remove that the names, the types and the tool outputs do not supply. On BFCL, the D2 effect flowed through a dozen parameters whose description held the only copy of an enum, a format or an entity type; ToolSandbox has no such parameter and no D2 effect. The one suggestive exception points the same way: Qwen3's −6.3 is concentrated in the 36 *canonicalization* scenarios (70.0 to 58.5; Table 13), in which the user states a date, a duration or a unit in surface form that the agent must convert to a POSIX timestamp or a unit name before calling, that is, in exactly the scenarios where a format convention has to be applied.

What D5 removes on ToolSandbox is the ability to tell 24 to 28 tools apart by anything but their argument structure, and much of that structure is shared: six tools take no arguments, eight take one, and the four `set_*_status` tools all take a single boolean. The behavioural signature of the collapse matches BFCL's. The number of scenarios in which the agent never calls a tool rises from 6, 3, 7 and 0 (Luna, Haiku, Qwen3, Gemma 4) to 27, 19, 16 and 34; the number that run to the 30-message cap rises from 2 to 23 for Luna and from 10 to 32 for Qwen3; and execution errors per scenario rise three- to five-fold for Luna, Qwen3 and Gemma 4. Two models mostly fall silent (Haiku, Gemma 4) and two mostly probe to the limit (Luna, Qwen3), the same split as in Section 4.3. Because the user is simulated, a silent or flailing agent is also

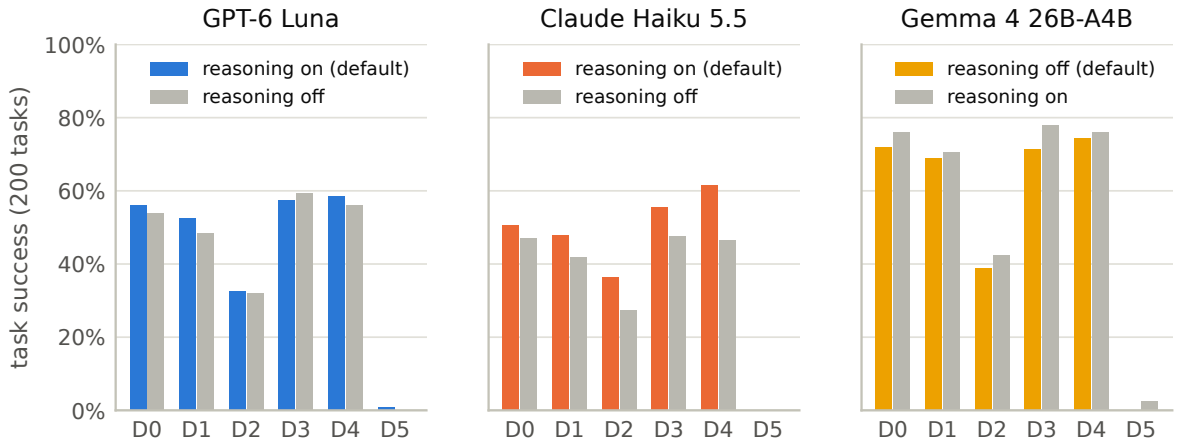


Figure 5: Task success by condition for the three reasoning-controlled pairs: the same model id with OpenRouter's reasoning switch explicitly off (Luna, Haiku) or on (Gemma 4), against the default run of the main study. The shape of the documentation effect is the same in every pair.

Table 6: Reasoning-controlled pairs on BFCL. *: the model's default, i.e. the main-study run. "Tokens" is the mean number of reasoning tokens per task at D0. ΔD2 and ΔD4 are the paired differences from that variant's own D0 (percentage points, 95% bootstrap CI, exact McNemar p).

			task success (%)									
Model	Reasoning	Tokens	D0	D1	D2	D3	D4	D5	ΔD2 [95% CI]	p	ΔD4 [95% CI]	p
GPT-6 Luna	on*	330	56.0	52.5	32.5	57.5	58.5	1.0	-23.5 [-30.5, -16.5]	<0.0001	+2.5 [-3.0, 8.0]	0.4869
	off	0	54.0	48.5	32.0	59.5	56.0	0.0	-22.0 [-30.0, -14.0]	<0.0001	+2.0 [-5.0, 8.5]	0.6655
Claude Haiku 5.5	on*	517	50.5	48.0	36.5	55.5	61.5	0.0	-14.0 [-21.0, -7.0]	0.0002	+11.0 [5.5, 16.5]	0.0003
	off	0	47.0	42.0	27.5	47.5	46.5	0.0	-19.5 [-26.0, -13.0]	<0.0001	-0.5 [-6.5, 5.5]	1.0000
Gemma 4 26B-A4B	off*	0	72.0	69.0	39.0	71.5	74.5	0.0	-33.0 [-40.0, -26.0]	<0.0001	+2.5 [-0.5, 6.0]	0.2266
	on	1110	76.0	70.5	42.5	78.0	76.0	2.5	-33.5 [-41.0, -26.5]	<0.0001	+0.0 [-2.5, 2.5]	1.0000

talked at for longer: the user simulator's token usage rises by 40% to 180% in D5.

Two features of ToolSandbox limit what can be read from it. Its score is a similarity rather than a verdict, and the 18 *insufficient-information* scenarios, in which the correct behaviour is to ask rather than act, are scored low for every model in every condition (24 to 74 at D0; Table 13) and move by less than a point under D2. And the user is an LLM: the same user model and the same seed were used in every condition, but the user is not deterministic, so each paired difference includes user noise as well as agent noise. The sign-flip tests account for this noise; they are not a substitute for a repeated run, which we did not make on ToolSandbox.

3.6 Reasoning-controlled pairs

Three of the four models can be run with the same weights and reasoning explicitly switched (Section 2): Luna and Haiku reason by default and were rerun with reasoning off, Gemma 4 does not and was rerun with reasoning on, each on all 200 tasks under D0 to D5. The switch took: with reasoning off, Luna and Haiku emitted no reasoning tokens in any of their 1 200 trajectories; with it on, Gemma 4 emitted 1 110 per task at D0. Figure 5 and Table 6 give the results.

Reasoning does not change the documentation effect. The D2 drop is −23.5 pp with Luna's reasoning on and −22.0 pp with it off, −14.0 and −19.5 pp for Haiku, −33.0 and −33.5 pp for Gemma 4; all six are significant at the corrected threshold, and the D3 null holds in every variant. D1 is again a few points negative in every variant (−5.5, −5.0 and −5.5 pp for Luna off, Haiku off and Gemma 4 on; $p = 0.16$, 0.15 and 0.05), so removing tool descriptions has now cost between 2.5 and 6 points in seven runs out of seven. D5 success is at most 2.5% in every variant; Gemma 4

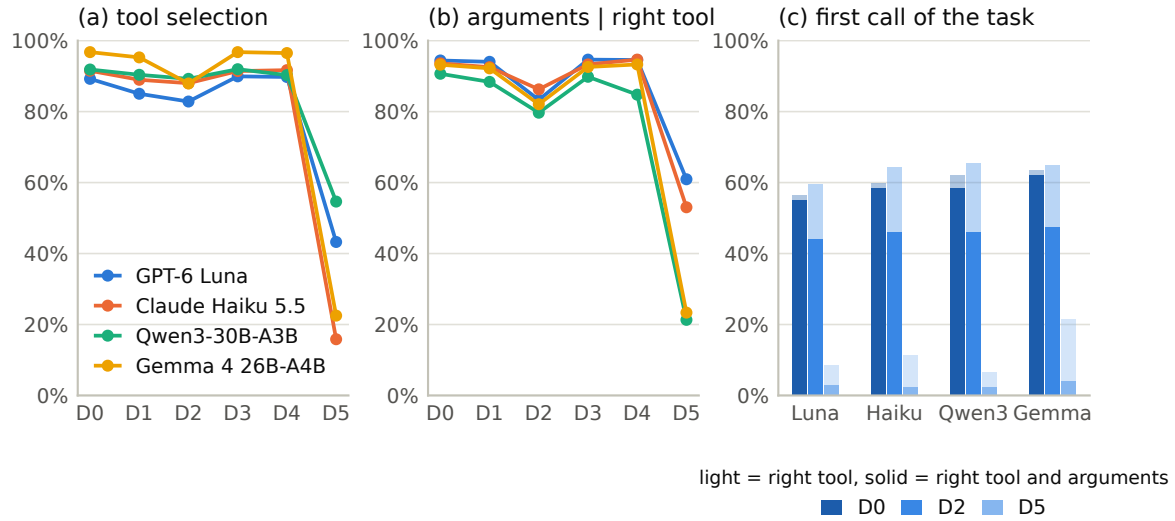


Figure 6: (a) Tool-selection accuracy: share of ground-truth calls whose tool the model invoked by the end of the required turn. (b) Argument accuracy conditional on the right tool: share of tool hits that were also exact-argument hits. (c) First call of the task: share of tasks whose first call used a ground-truth tool (light) and used it with exactly the ground-truth arguments (solid), for D0, D2 and D5.

with reasoning on passes five D5 tasks, the most of any run, at 8 659 reasoning tokens per task.

What reasoning changes is the level and, for one model, the D4 result. Turning reasoning off costs Luna nothing measurable at any condition (paired differences of -4.0 to $+2.0$ pp, $p \geq 0.29$). It costs Haiku between 3.5 and 15.0 pp at every condition, significantly at D2 (-9.0 pp, $p = 0.0009$) and D4 (-15.0 pp, $p = 0.0001$). Haiku's D4 gain of Section 3.3, $+11.0$ pp with reasoning on, is -0.5 pp with it off ($p = 1.0$): whatever produces it belongs to Haiku's reasoning mode, not to the generic parameter names, and it is no longer a candidate for a documentation effect. Turning reasoning on raises Gemma 4 by 1.5 to 6.5 pp at every condition but D5, nominally significant only at D3 ($+6.5$ pp, $p = 0.024$), and leaves its D2 drop where it was.

The reasoning budget responds to the interface even though success does not: per task, Luna spends 330 reasoning tokens at D0, 390 at D2 and 2 294 at D5; Haiku 517, 559 and 1 258; Gemma 4 with reasoning on 1 110, 2 554 and 8 659. The models think longer when the documentation says less, and recover none of the loss by doing so.

4 Analysis: where documentation matters

The aggregate results say that parameter descriptions matter. This section uses the recorded trajectories to ask where in the tool-use process they matter: when the model selects a tool, when it constructs the arguments, or when it recovers from an error. It closes with the two follow-up conditions that test the answer.

4.1 Selecting tools versus constructing arguments

The analysis plan asked whether documentation helps the model pick the right tool or fill in the right arguments. Figure 6 and Table 7 separate the two. Under the single ablations tool-selection accuracy is nearly flat: between D0 and D2 it moves by 2.6 to 8.8 points, and part of even that movement is a cascade effect, because a turn derailed by a wrong argument may never reach its later ground-truth tools. Argument accuracy given the right tool drops in D2 for every model: 94.4% → 83.3% (Luna), 93.6% → 86.3% (Haiku), 90.7% → 79.7% (Qwen3) and 93.2% → 82.1% (Gemma 4).

Table 7: Tool selection versus argument construction (%). "Sel." is tool-selection accuracy, "Arg." argument accuracy given the right tool, "1st tool" the share of tasks whose first call used a ground-truth tool, and "1st exact" the share whose first call matched a ground-truth call exactly.

	D0				D2				D5			
Model	Sel.	Arg.	1st tool	1st exact	Sel.	Arg.	1st tool	1st exact	Sel.	Arg.	1st tool	1st exact
GPT-6 Luna	89.2	94.4	56.5	55.0	82.8	83.3	59.5	44.0	43.3	60.9	8.5	3.0
Claude Haiku 5.5	91.5	93.6	60.0	58.5	88.0	86.3	64.5	46.0	15.8	53.0	11.5	2.5
Qwen3-30B-A3B	91.9	90.7	62.0	58.5	89.2	79.7	65.5	46.0	54.6	21.3	6.5	2.5
Gemma 4 26B-A4B	96.8	93.2	63.5	62.0	87.9	82.1	65.0	47.5	22.5	23.3	21.5	4.0

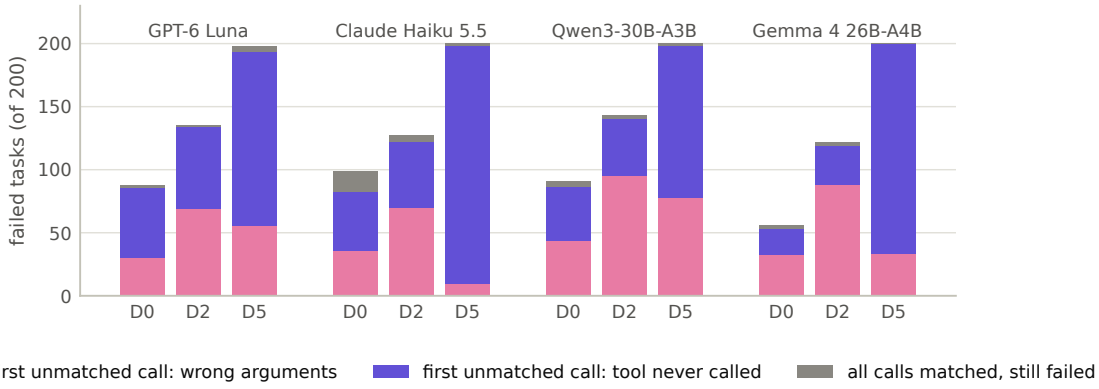


Figure 7: Earliest divergence in failed tasks. For each failed task, the first ground-truth call (in turn order) without an exact match is classified by whether the tool was called with wrong arguments or never called at all; "all calls matched" means every ground-truth call was matched at some point but the task still failed, usually because of extra state-changing calls or a call made in a later turn than required.

The first call of each task, which is free of cascade effects, tells the story most cleanly (Figure 6c). In D2 the right tool is chosen for the first call slightly *more* often than in D0 for every model (56.5% → 59.5%, 60.0% → 64.5%, 62.0% → 65.5%, 63.5% → 65.0%), but the right tool with the right arguments falls by 11.0, 12.5, 12.5 and 14.5 points. The models know *what* to call; they no longer know what to put in it.

Figure 7 (and Table 14 in the appendix) look at the same question from the failure side. Among D2 failures the earliest divergence is a wrong-argument call rather than a missing tool for 69 of 135 (Luna), 70 of 127 (Haiku), 95 of 143 (Qwen3) and 88 of 122 (Gemma 4) tasks. Compared with D0, the number of failures that begin with a wrong-argument call rises by 39, 34, 51 and 55, whereas the number that begin with a never-called tool rises by only 9, 5, 2 and 11. D5 is different in kind: tool selection itself collapses (43.3%, 15.8%, 54.6%, 22.5%) and most failures begin with a tool that was never called.

4.2 Which descriptions matter

Table 8 links each new argument error to the description that was removed. Six parameters account for most of the new errors, and they are the same six for every model. What their descriptions have in common is that they carried information the schema did not express structurally:

- an allowed-value list written only in prose: `place_order.order_type` ("Buy/Sell") and `startEngine.ignitionMode` ("[Enum]: START, STOP"), both plain string fields with no schema `enum`;
- an entity type that differs from what the parameter name suggests: `estimate_distance.cityA` and `cityB` take zip codes, not city names;

Table 8: Parameters whose values go wrong when their description is removed. For every ground-truth call matched by the right tool but wrong arguments, the closest same-tool call was found and the differing parameters recorded. Counts are pooled over the four models (800 task runs per condition), sorted by the D2−D0 difference; per-model D2 counts on the right. The second column quotes the description that D2 removed.

Tool.parameter	Removed description	D2	D0	Luna	Haiku	Qwen3	Gemma
`place_order.order_type`	*Type of the order (Buy/Sell).*	**108**	0	26	27	27	28
`startEngine.ignitionMode`	*The ignition mode of the vehicle. [Enum]: ["START", "STOP"]*	**97**	0	25	14	28	30
`estimate_distance.cityA`	*The zipcode of the first city.*	**75**	4	20	20	16	19
`estimate_distance.cityB`	*The zipcode of the second city.*	**75**	4	20	20	16	19
`post_tweet.tags`	*List of tags for the tweet. Tag name should start with #. This is only relevant if the user wants to add tags to the tweet.*	**72**	3	22	18	12	20
`post_tweet.mentions`	*List of users mentioned in the tweet. Mention name should start with @. This is only relevant if the user wants to add mentions to the tweet.*	**45**	4	15	11	8	11
`post_tweet.content`	*Content of the tweet.*	**37**	26	6	17	3	11
`get_zipcode_based_on_city.city`	*The name of the city.*	**10**	0	1	4	4	1
`compute_exchange_rate.base_currency`	*The base currency. [Enum]: USD, RMB, EUR, JPY, GBP, CAD, AUD, INR, RUB, BRL, MXN*	**8**	0	3	1	2	2
`get_available_stocks.sector`	*The sector to retrieve stocks from (e.g., 'Technology').*	**8**	0	3	0	4	1

- a format convention: `post_tweet.tags` must start with `#` and `mentions` with `@`.

Without the description, models wrote `"buy"` instead of `"Buy"`, `"start"` instead of `"START"`, passed city names where zip codes were required, or folded the hashtag into the tweet body instead of the `tags` list (the `content` row in the table is the mirror image of `tags`: when the hashtag is folded into the content, the content value is also wrong). The last pattern appears in three of the four case studies in Section 4.5. The backends react to these errors in two different ways. `place_order` lower-cases `order_type` for its funds check but stores the value as given, so `"buy"` is accepted silently and only the final-state comparison catches it; `post_tweet` likewise stores the hashtag-laden content with an empty tag list. `startEngine("start")` returns "Invalid ignition mode." and `estimate_distance` with city names returns "distance not found in database.", so for those two parameters the model received an explicit error and, as Section 4.3 shows, usually did not repair the argument.

Task-level attribution. Table 8 counts argument errors, not tasks, so it does not by itself establish how much of the D2 drop the six parameters explain. The ground truth of 91 of the 200 tasks contains at least one of them. Of the tasks that pass in D0 and fail in D2, 44 of 53 (Luna), 37 of 42 (Haiku), 50 of 63 (Qwen3) and 64 of 71 (Gemma 4) are among those 91. Splitting the success rates accordingly (Table 9), the D2 drop on the 91 tasks is 47.3, 36.3, 51.6 and 70.3 pp, while on the other 109 tasks it is +3.7, −4.6, +4.6 and +1.8 pp, within the run-to-run variation measured in Section 3.3. At task-level resolution the D2 effect is an effect on tasks that use these six parameters; Section 4.6 shows that a tail of other prose-only enums adds a few points on top.

This narrows the headline finding. Parameter descriptions matter here less as "meaning" and more as the place where this benchmark keeps its enums, formats and entity conventions. Section 4.6 tests whether a schema that moves those into `enum`, `pattern` and precise parameter names recovers the D2 loss with the descriptions still blank.

Table 9: Success (%) on the 91 tasks whose ground truth uses at least one of the six parameters of Table 8 ("six") and on the other 109 tasks ("other"), per condition. The first column counts the tasks that pass in D0 and fail in D2 and, of those, how many are six-parameter tasks.

	D0-only tasks	D0		D2		D6		D7	
Model	six / all	six	other	six	other	six	other	six	other
GPT-6 Luna	44/53	51.6	59.6	4.4	56.0	50.5	56.9	49.5	54.1
Claude Haiku 5.5	37/42	46.2	54.1	9.9	58.7	49.5	57.8	51.6	57.8
Qwen3-30B-A3B	50/63	60.4	49.5	8.8	45.0	59.3	44.0	49.5	40.4
Gemma 4 26B-A4B	64/71	75.8	68.8	5.5	67.0	75.8	65.1	64.8	64.2

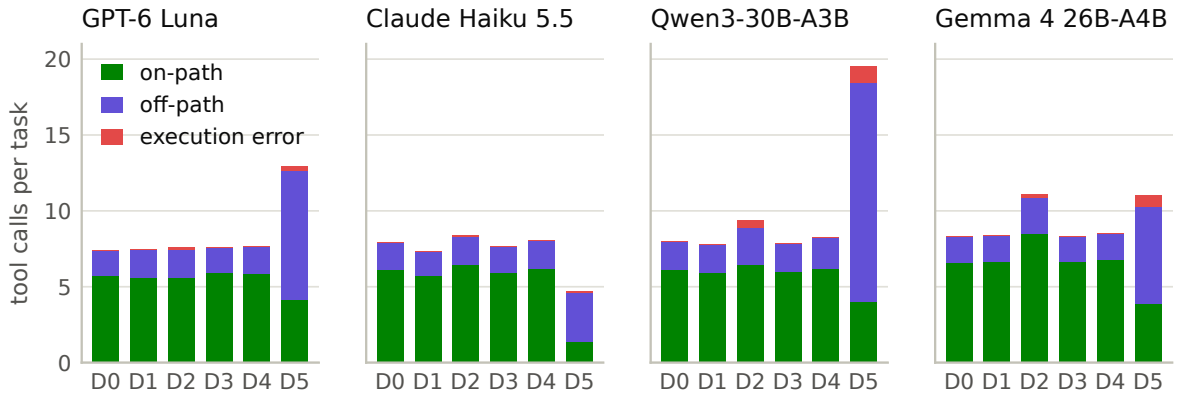


Figure 8: Composition of tool calls per task (mean over all 200 tasks, failures included): on-path calls, off-path calls that executed without error, and calls that returned an execution error.

4.3 Tool-use behaviour, recovery and exploration

Does a model compensate for missing documentation by interacting more? Figure 8 (full numbers in Table 16) says that under the single ablations it does not. D2 adds 0.2, 0.5, 1.4 and 2.8 calls per task over D0 for Luna, Haiku, Qwen3 and Gemma 4 respectively. The exception is Gemma 4, which repeats calls when it is unsure (3.39 repeated calls per task in D2 against 1.24 in D0) and hits the 20-step limit in 20 tasks (4 in D0). The composition shifts rather than the count: failed calls per task rise from at most 0.02 in D0 to between 0.13 and 0.51 in D2.

D5 produces two opposite behaviours. Haiku makes *fewer* calls than in D0 (4.7 against 7.9), does not call any tool in the first turn of 34.5% of tasks, and most of the calls it does make are off-path. Qwen3 makes 19.5 calls per task, touching 15.6 distinct tools of the roughly 28 available, with 1.09 failed calls per task and an execution error in 51% of tasks. Luna sits between (12.9 calls, 9.1 distinct tools, no first-turn call in 22% of tasks), as does Gemma 4 (11.0 calls, 5.0 distinct tools). The ECDFs in Figure 9 show that D2 shifts the call distribution only slightly to the right, while D5 pulls it in opposite directions by model: Haiku's curve rises steeply at 0 to 3 calls, Qwen3's stretches out to the step cap.

Neither restraint nor probing turns into recovered tasks (Figure 10, left; Table 15). In D2, of the tasks that contained an execution error, 12% of 24 (Luna), 22% of 23 (Haiku), 20% of 30 (Qwen3) and 21% of 29 (Gemma 4) still passed. D0 recovery rates look higher but rest on one to four tasks each. In D5 no model recovered a single task after a failed call: 0 of 38, 17, 102 and 30. The efficiency frontier (Figure 10, right) is flat across single ablations: D0 through D4 sit at 7 to 9 calls per task for every model and the vertical spread between them is the D2 effect. D5 moves points right (Qwen3, Luna, Gemma 4) or left (Haiku) without moving them up.

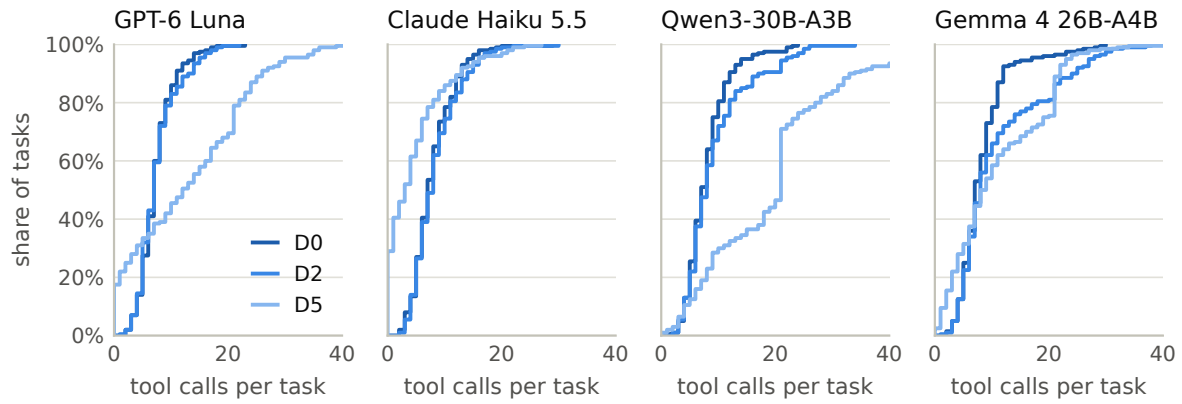


Figure 9: Empirical cumulative distribution of tool calls per task for D0, D2 and D5. All 200 tasks per curve, including failed and force-terminated runs. A curve further to the right means more calls.

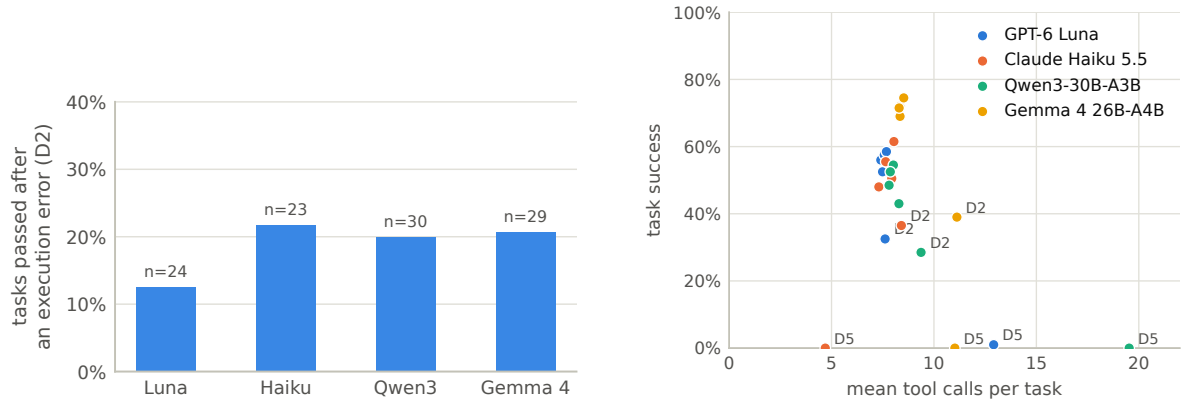


Figure 10: Left: recovery after an execution error in D2: share of tasks containing at least one failed call that still passed, with the number of such tasks. D0 denominators are 1 to 4 tasks and are not comparable; in D5 no model recovered any of its 17 to 102 such tasks (Table 15). **Right:** efficiency frontier, task success against mean tool calls per task, one point per model and condition; D2 and D5 points are labelled.

4.4 Failure outcomes

BFCL assigns one terminal error per failed task (Figure 11, Table 17). In D2 the extra failures are almost all "wrong final state" (61 → 96, 64 → 86, 64 → 97 and 35 → 76 tasks from D0 to D2): the model acted, but on the wrong values. In D5 the mix changes with the model's strategy. Haiku's and Luna's D5 failures are dominated by turns that ended without any tool call (78 and 50 tasks), Qwen3's by hitting the 20-step limit (57 tasks) and Gemma 4's by a wrong final state after probing (94 tasks).

4.5 Case studies

Case studies were selected by a rule applied identically to every model: among tasks that pass in D0 and fail in D2, take the one with the fewest ground-truth calls (lowest task id on ties), and show the same task in D0, D2 and D5. The rule favours short tasks so that the whole trajectory can be shown; it does not select for drama. Figure 1 shows the Claude Haiku case, the other three are summarised below, and Appendix B gives every call of all four cases under the three conditions.

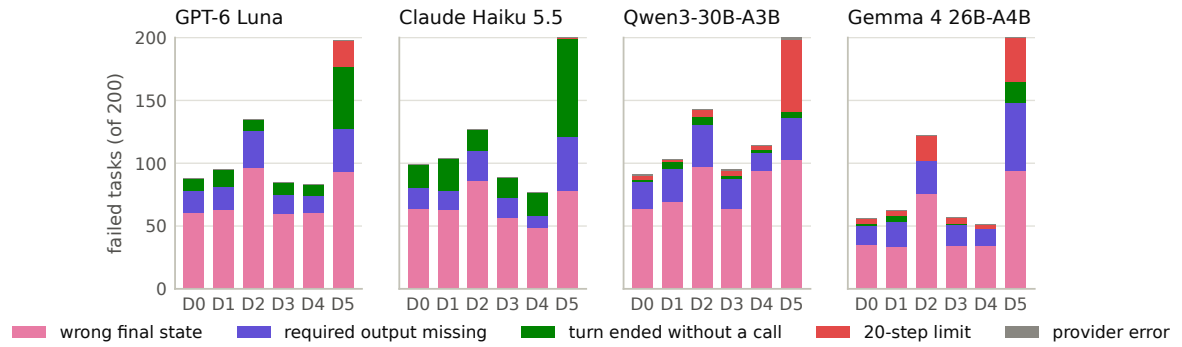


Figure 11: BFCL's terminal error per failed task, by condition. These are observed outcomes, not causes: a wrong final state can come from a wrong tool, a wrong argument or a missed step.

Claude Haiku 5.5, task 78 (Figure 1). Turn 0 is identical in D0 and D2. In turn 1, D0 passes the hashtag as `tags=["#BusinessOnTheMove"]`; D2 folds it into `content`, so the backend records a tweet with no tags and the final state differs. The only information that changed between the two runs is the removed `tags` description quoted in the figure. In D5 the model never calls a tool and tells the user it will not guess which of the unlabeled tools posts to social media; that is a refusal, not an exploration.

Gemma 4 26B-A4B, task 11. The user asks to list the files in a `/temp` directory (which does not exist; the files are in the current directory) and then to tweet their names with the hashtag `#fileshowcase`. Both D0 and D2 wander in turn 0, repeating `ls` and `find` until they settle, and both end turn 0 in the correct state. The outcome is decided in turn 1 by the same hashtag-in-content error as the Haiku case. In D5 the single call is `get_user_stats(username="ls -a /temp")`: the model picked a Twitter tool and passed the shell command it wanted to run as its argument.

GPT-6 Luna, task 177. The user asks to book a business-class flight from Los Angeles to Tokyo and then to "confirm the booking's details". D2 shows a recovered argument error and an unrecovered tool choice. In turn 0 it first calls `get_flight_cost` with city names (the removed descriptions said the parameters take airport codes), gets an execution error, corrects to `LAX` and `HND`, and books correctly. In turn 1, D0 calls `retrieve_invoice` while D2 calls `get_booking_history`. Both return the booking, but BFCL requires the invoice output, so D2 fails with "required output missing". In D5 the model answers turn 0 with an essay about verifying itineraries and never calls a tool.

Qwen3-30B-A3B, task 42. The user asks to create `Notes2023.txt` in an existing `Lectures` folder, write a sentence to it, and count its characters. The divergence is in turn 0. In D0, after `cd Lectures` fails, the model calls `pwd`, discovers it is already inside `/Lectures`, and creates the file. In D2 it stops after the failed `cd` and an empty `ls`, so turn 0 ends with no file created; in turn 1 it creates a new `Lectures` folder inside the existing one and writes the file there. Every later call succeeds, but the directory tree has an extra folder and BFCL scores a wrong final state. Nothing about the documentation of the tools involved differs between D0 and D2; this is a case where the run diverged on a near-tie decision and the evaluator is strict about state. It is a reminder that not every D0/D2 discordant task is a documentation effect, which is why the paired statistics count both discordant cells. In D5 the model tries eleven different tools with guessed arguments and gives up.

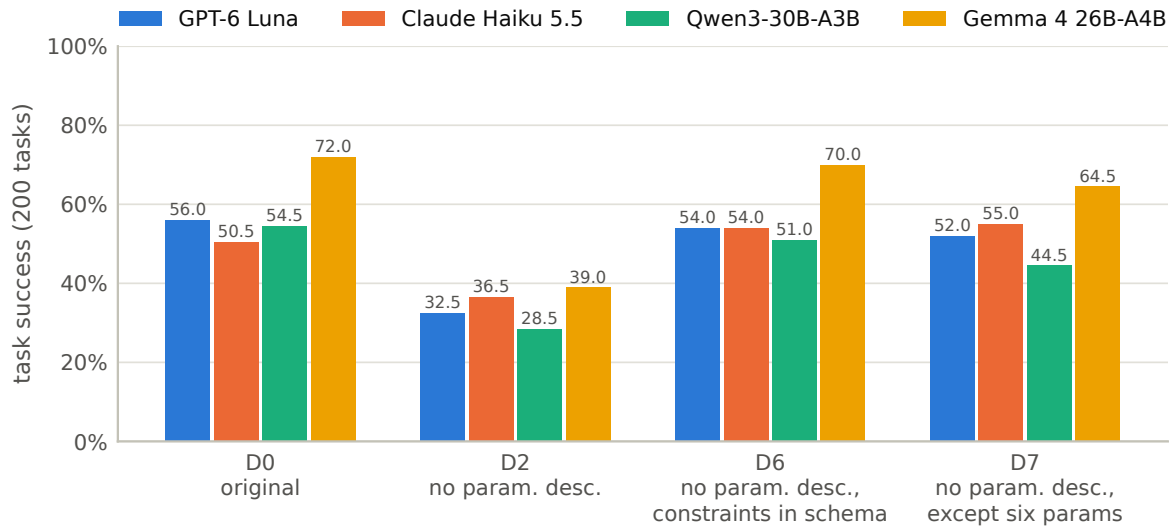


Figure 12: Follow-up conditions. D6 and D7 both blank every parameter description, as D2 does. D6 moves the prose-only constraints into the schema (`enum`, `pattern`, two renamed parameters); D7 keeps the descriptions of the six parameters of Table 8 only. Same 200 tasks as every other bar.

4.6 Follow-up: restoring the structure without the prose

Section 4.2 argued that the parameter descriptions that matter are the ones carrying constraints the schema could have expressed. Figure 12 and Table 10 test that claim with the two conditions defined in Section 2. In D6, with every description still blank but the prose-only enums moved into `enum`, patterns added for tags and mentions, and the two zip-code parameters renamed, success is 54.0% (Luna), 54.0% (Haiku), 51.0% (Qwen3) and 70.0% (Gemma 4), against 56.0, 50.5, 54.5 and 72.0% in D0: every model is within 3.5 points of its baseline, no D6–D0 difference is significant, and every D6–D2 difference is, recovering 21.5, 17.5, 22.5 and 31.0 points. D7, which restores only the six descriptions, reaches 52.0, 55.0, 44.5 and 64.5%: it recovers most of the D2 loss for every model but stays 4 to 10 points below D0 for three of them (significant for Qwen3 at the uncorrected level), which is what one expects if a few further prose-only constraints outside the six contribute a little each. Table 9 shows the same picture split by task: on the 91 tasks that use the six parameters, D6 is back at D0 for every model, and on the other 109 tasks D2 and D6 stay within the noise band of D0 for every model, while D7 is slightly below D0 for Qwen3 (-9 points). The parameters that go wrong in D7 but not in D6 are exactly the remaining prose-only enums that D6 converts and D7 does not, led by `lockDoors.door` (20 wrong values pooled over the models against 3 in D0) and `activateParkingBrake.mode` (7 against 0).

Was the constraint read or enforced? A provider may enforce an `enum` during decoding, in which case part of the D6 recovery would belong to the decoder rather than the model. The trajectories allow a partial check: counting every value a model emitted for the 13 constrained parameters (for the two `pattern` parameters, "off-constraint" means a present list item that fails the regex, not an omitted tag, so the D0 `tags` errors of Table 8, which are omissions, do not appear here), the share outside the constraint is 37 to 69% in D2 for all four models (`ignitionMode="start"`, `order_type="buy"`, `door="all"`) and 0.0% for Luna, Haiku and Gemma 4 in D6, with 3 of 455 values (0.7%) off-constraint for Qwen3. The Qwen3 values show that at least its providers did not enforce the enum, and the fact that the same three models were also at 0.0% in D0, where the constraint existed only as prose that no decoder could enforce, shows that they follow a shown constraint whether or not it is enforced. The check cannot rule out that Azure or Google additionally enforced the schema in D6; what it rules out

Table 10: Follow-up conditions against D0 and against D2 on the same 200 tasks: success of the base and follow-up condition, the discordant cells (tasks passing only in the base condition, only in the follow-up), the paired difference with its 95% bootstrap interval, and the exact McNemar p.

Model	Comparison	base %	f.-up %	base only	f.-up only	Δ pp	95% CI	p
GPT-6 Luna	D6 vs D0	56.0	54.0	19	15	-2.0	[-7.5, 4.0]	0.6076
	D6 vs D2	32.5	54.0	6	49	+21.5	[15.0, 28.0]	<0.0001
	D7 vs D0	56.0	52.0	23	15	-4.0	[-10.0, 2.0]	0.2559
	D7 vs D2	32.5	52.0	7	46	+19.5	[13.0, 26.0]	<0.0001
Claude Haiku 5.5	D6 vs D0	50.5	54.0	14	21	+3.5	[-2.0, 9.5]	0.3105
	D6 vs D2	36.5	54.0	7	42	+17.5	[11.0, 24.0]	<0.0001
	D7 vs D0	50.5	55.0	16	25	+4.5	[-2.0, 11.0]	0.2110
	D7 vs D2	36.5	55.0	13	50	+18.5	[11.0, 25.5]	<0.0001
Qwen3-30B-A3B	D6 vs D0	54.5	51.0	23	16	-3.5	[-10.0, 2.5]	0.3368
	D6 vs D2	28.5	51.0	6	51	+22.5	[16.0, 29.5]	<0.0001
	D7 vs D0	54.5	44.5	32	12	-10.0	[-16.5, -3.5]	0.0037
	D7 vs D2	28.5	44.5	14	46	+16.0	[9.0, 23.5]	<0.0001
Gemma 4 26B-A4B	D6 vs D0	72.0	70.0	13	9	-2.0	[-6.5, 2.5]	0.5235
	D6 vs D2	39.0	70.0	9	71	+31.0	[23.5, 39.0]	<0.0001
	D7 vs D0	72.0	64.5	25	10	-7.5	[-13.0, -2.0]	0.0167
	D7 vs D2	39.0	64.5	7	58	+25.5	[18.5, 32.5]	<0.0001

is that enforcement is needed to explain the recovery. The distinction matters for the audit recommendation in Section 5, because enforcement is not free: Ray (2026) reports that for a 1.5B-parameter model on a calendar tool-call task, hard schema decoding kept every output schema-valid while executable accuracy fell from 91.5% to 48.0%, so "constraint shown" and "constraint enforced" should be chosen separately.

Two caveats attach to D6. First, it relies on the serving stack passing `enum` and `pattern` through to the model; OpenRouter did so for all four providers here (the schemas the models received are stored with each trajectory), but BFCL itself strips several constraint fields for Gemini-style endpoints and other stacks may do the same. Second, D6 was designed with knowledge of which parameters failed in D2, so it tests the specific hypothesis of Section 4.2, not the general proposition that any schema can be made prose-free.

5 Discussion

What the results support. On this benchmark, parameter descriptions are the documentation component the models depend on; the dependence is at the argument-construction step rather than tool selection; it is carried by constraints that the schema did not encode, since expressing them structurally restores baseline success with the prose gone; and removing all four components removes the ability to do the tasks at all, for every model. These conclusions rest on paired comparisons over the same 200 tasks, replicate across four models from four providers with different architectures and training regimes, are corroborated by three independent views of the trajectories (first-call accuracy, earliest divergence and per-parameter error linkage) that do not depend on BFCL's verdict, and in the case of the structural claim were confirmed by a follow-up condition designed to falsify it. The claim also makes a prediction for a benchmark whose descriptions hold no such constraints: no D2 effect. ToolSandbox is such a benchmark, and showed none (Section 3.5). And the effect is not a property of a reasoning mode: it is the same size with reasoning on or off for the three models that can be switched (Section 3.6).

Descriptions as a carrier of structure. The most actionable finding is the pair of Table 8 and Table 10. The descriptions whose removal hurt were not explaining what a parameter *means*;

a model does not need prose to understand that `order_type` is the type of the order. They were carrying an allowed-value list, a format rule or an entity convention that JSON Schema could have expressed as `enum`, `pattern` or a better-chosen name, and that the benchmark's authors happened to put in prose. When those constraints are expressed structurally (D6), every model performs at baseline with no parameter descriptions at all. For tool designers this suggests a simple audit: for every parameter description, ask whether it contains information that is not in the schema's structural fields, and if so move it there, keeping the prose for what structure cannot say. The audit presupposes that the serving stack preserves the structural fields; where it does not, the prose is the only channel and D2 is the relevant condition. The D4 result points the same way from the other side: for three of the four models, replacing a parameter name with `arg_N` while *keeping* its description cost nothing. Haiku's apparent gain on D4 is diffuse and shrinks to a non-significant +5.0 pp ($p = 0.13$) against the repeated D0 run (Section 3.3), so we treat it as suggestive at most; Qwen3's −11.5 pp on D4 is well outside the noise floor and shows that the dependence on names is model-specific. Note that the frontier and open-weight pairs also differ in reasoning (Section 2), so this split cannot be attributed to model family.

Tool names and tool descriptions. The small D1 and null D3 results should not be over-read. The tools in BFCL's multi-turn base category have descriptive names (`post_tweet`, `book_flight`), the user prompts name the actions in plain language, and the tool outputs are fully informative; with 18 to 39 tools per task, a model can also distinguish tools by their argument structure. Under those conditions the tool description is redundant with the name, and the name is redundant with the parameters, so removing either one leaves the other. D5, where all redundancy is removed at once, shows that the information was being used; it was simply available from more than one place. ToolSandbox, whose 24 to 28 tools include fourteen that take at most one argument, shows D1 differences of mixed sign, the same D3 null for three models and a nominal 6-point D3 loss for Gemma 4, a first sign of the name mattering once argument structure is less distinctive. In settings with ambiguous or overlapping tools (the candidate-set confusion studied by Chattopadhyay et al. (2026) and Rabinovich and Anaby-Tavor (2025)) we would expect D1 and D3 to matter more.

Exploration does not substitute for documentation. The experiment design asked whether models compensate for missing documentation through multi-turn interaction. They do not, on this benchmark, and the two strategies observed in D5 fail for different reasons. Declining to act (Haiku, Luna) fails immediately because the evaluator requires the state change. Probing (Qwen3, Gemma 4) fails because BFCL caps a turn at 20 steps and scores the final state, so every exploratory call that mutates state is penalised and every one that does not consumes budget. Recovery after an execution error is poor even in D2 (12 to 22%), which suggests that models treat an execution error as a signal to try a different tool or give up rather than to re-read the schema and change the offending argument. Whether a harness that allowed dry runs, more steps or explicit schema introspection would change this is an open question.

Implications for benchmarks. The D2 effect flows through a dozen parameters in three backends, six of them dominant (Section 4.2), and disappears when their constraints are written into the schema (Section 4.6). A benchmark that encoded its enums and formats structurally would be more robust to documentation quality and would isolate the models' understanding of task semantics from their ability to parse prose. Conversely, the fact that frontier models fail on `"buy"` versus `"Buy"` when the prose is removed shows how literally stateful benchmarks score: BFCL's single-turn categories compare argument strings after lower-casing and stripping punctuation, but the multi-turn state checker compares backend attributes exactly,[1] so a value

[1] At the pinned commit, `standardize_string` in `bfcl_eval/eval_checker/ast_eval/ast_checker.py` removes spaces and punctuation and lower-cases both sides before the single-turn comparison, whereas

the single-turn scorer would accept fails the stateful one. How much of a reported function-calling score rests on such conventions rather than on reasoning is worth knowing for any benchmark.

6 Related work

Tool learning and function calling. Teaching language models to call external tools has progressed from self-supervised insertion of API calls (Schick et al., 2023) and reasoning-and-acting prompting (Yao et al., 2023) to instruction-tuned models trained on tens of thousands of real APIs (Patil et al., 2023; Qin et al., 2024). Qu et al. (2025b) survey the area and decompose tool use into task planning, tool selection, tool calling and response generation; our decomposition into tool selection and argument construction follows their framing, with the two stages measured separately against ground truth.

Benchmarks. API-Bank (Li et al., 2023) and the single-turn categories of BFCL (Patil et al., 2025; Yan et al., 2024) evaluate individual calls by abstract-syntax-tree or execution matching. Stateful, multi-turn evaluation was introduced by BFCL V3 (Mao et al., 2024), τ-bench (Yao et al., 2025) and ToolSandbox (Lu et al., 2024), which score a trajectory by comparing the final backend state (and required outputs) with a ground truth rather than by matching individual calls. We use BFCL's multi-turn base category because its state-based scorer, deterministic backends and per-turn ground truth allow a documentation change to be isolated from every other variable.

Tool documentation. Hsieh et al. (2023) showed that documentation alone can replace few-shot demonstrations for zero-shot tool use. EasyTool (Yuan et al., 2025) rewrites documentation into concise, standardised instructions, DRAFT (Qu et al., 2025a) refines documentation from execution feedback, and Guo et al. (2026) learn to rewrite descriptions for trace-free deployment, transferring supervision from settings with execution traces to ones without; in an appendix, the last of these report having to correct StableToolBench parameter schemas whose required fields, types and enums did not match the servers, a second instance of a benchmark keeping its constraints somewhere other than the schema. These works treat the documentation as a single object to be improved; we instead ask which *component* of it carries the value, and find that on a stateful benchmark it is the parameter descriptions, and within those the enums, formats and entity conventions that the schema does not otherwise encode. A small pre-registered ablation on Model Context Protocol tools (Clouâtre, 2026) found that parameter detail can move from the tool-description string into per-parameter schema fields with no measurable loss, the same redundancy between tool-level and parameter-level prose that our D1 result shows.

Robustness of function calling. Rabinovich and Anaby-Tavor (2025) perturb BFCL queries and expand toolkits with semantically related distractors and find all evaluated models degrade. ToolRobustBench (Zheng et al., 2026) and AgentPerturb (Xu et al., 2026) study perturbations at several stages of a tool-calling pipeline, including rewriting tool descriptions, with mixed conclusions about how damaging interface perturbations are. An independent group's ablation comes closest to our null results from the other direction: Chattopadhyay et al. (2026) find that minor description rewording and semantics-preserving parameter renames do not meaningfully degrade accuracy, while distractor tools with overlapping descriptions do, which is our D1/D3 finding together with our expectation that D1 and D3 would matter more with overlapping tools. None of these works removes a documentation component outright while holding the rest of the benchmark fixed, which is the manipulation studied here.

`_compare_instances` in `bfcl_eval/eval_checker/multi_turn_eval/multi_turn_checker.py` tests every public attribute of the model's backend instance against the ground-truth instance with Python equality, and the multi-turn response checker likewise matches serialised outputs without normalisation.

Interface specification and structure. Sigdel and Baral (2026) vary only the form of the tool interface, free-form documentation against JSON Schema against schema plus structured validation diagnostics, with identical tool semantics across conditions, and find in a small pilot that schema conditions reduce interface misuse but not semantic misuse. Their design is the closest to our D6, and their third condition, which feeds the model a structured diagnosis of what was wrong with its call, is the intervention our recovery results leave open. Johnson et al. (2025) argue the opposite direction for tool *selection*: replacing structured function calling with a natural-language selection step improved accuracy across ten models. The two findings are compatible with ours: structure helps for argument *constraints*, which is where our effect lives, and may hurt at the selection step, which our ablations barely touch. On the cost side of structure, Ray (2026) shows for small models that enforcing a schema during decoding can keep outputs valid while lowering executable accuracy, so the recommendation of Section 5 is to express constraints in the schema, not necessarily to enforce them.

7 Limitations

- **Two benchmarks, mostly one run each.** The main results are on BFCL's 200 multi-turn base tasks over eight synthetic backends, run once per condition at temperature 0.001, with a single repeated D0 run for one model; the ToolSandbox replication covers 91 scenarios, run once, with an LLM-simulated user whose own variation enters every paired difference. That repeat (Section 3.3) shows two identical runs disagreeing on tens of tasks, so differences of a few points, including the D1 cost and Haiku's D4 gain, cannot be resolved by this design. The D2, D5 and D6-versus-D2 effects are an order of magnitude larger than the measured floor.
- **D5 is not zero semantics.** Types, requiredness, enums, response-schema structure, user prompts and tool outputs all remain. The D5 collapse measures the joint contribution of names and descriptions given everything else, not what a model can do with no information.
- **Exact matching is strict.** Argument accuracy, exact hits and the first-divergence analysis treat semantically equivalent but differently spelled values as misses. This is applied equally to all conditions but makes the absolute accuracies conservative, and BFCL's own scorer is similarly strict about state.
- **D6 is a targeted test.** It was designed after the main analysis, with knowledge of which parameters failed, and it depends on providers passing `enum` and `pattern` through to the model, which they did here. It shows that structure can replace the prose on this benchmark; it does not show how a model would fare on an unfamiliar API whose constraints had never been written anywhere.
- **Provider and reasoning settings.** Models were accessed through OpenRouter with default settings. The reasoning-controlled pairs (Section 3.6) set only the on/off switch, not an effort level, and Qwen3 has no such switch, so its reasoning-free result has no reasoning counterpart. Latency is not compared because provider routing differs per model.
- **Evaluator strictness as a confound.** The Qwen3 case study shows a D0/D2 discordance with no documentation cause. The paired statistics account for such noise, but individual discordant tasks should not be read as documentation effects without inspecting the trajectory. The D5 result in particular is partly a property of BFCL's evaluator, which caps a turn at 20 steps and scores the final state, so that every exploratory call either consumes budget or mutates state; it should not be read as a claim about what the models could learn with a more permissive harness.
- **Two-provider deployment for Qwen3.** A third of Qwen3's trajectories had at least one request routed to a second provider, and success on those trajectories is lower in every

condition (Section 2). Qwen3's absolute numbers therefore describe a mixed deployment; its within-model differences are present on both subsets.

- **Provider errors.** One Qwen3 task (`multi_turn_base_44`) was rejected with an HTTP 400 by the provider in D0 and again in D6, so its failures are deterministic rather than transient; two other Qwen3 tasks failed once each. All are counted as failures, and dropping them pairwise changes no conclusion.

8 Conclusion

Holding everything in a stateful multi-turn function-calling benchmark fixed except the documentation shown to the model, four models from four providers depend on the same component, parameter descriptions, at the same step, argument construction, and for the same six parameters plus a small tail of other prose-only enums. What those descriptions supplied was not meaning but constraints: when the same constraints are written into the schema as enums, patterns and precise names, the models perform at baseline with every description blank. Tool names were free to remove here and tool descriptions nearly so, and no model compensated for missing documentation by exploring. A second benchmark whose descriptions carry no such constraints, ToolSandbox, showed no parameter-description effect at all and the same collapse without names and descriptions, which is what the mechanism predicts; and switching reasoning on or off moved the level of success but not the effect. The message for tool designers is to put structure in structural fields; the message for benchmark designers is that a function-calling score can rest on documentation conventions that have nothing to do with the task.

Reproducibility statement

The harness, the condition definitions (including the exact schema transformation applied in D6), the analysis code and the paper sources are at `https://github.com/Eazydraft/exp-tool-semantics-ablation`; BFCL and ToolSandbox are included as submodules pinned to the commits used. Every number, table and figure in this paper is computed from the trajectory files, each of which holds the task, the exact tool schemas sent to the model, the name maps, every raw request and response (with provider and token usage), BFCL's per-turn result and backend state, and the evaluator's verdict. The 6 400 trajectory files of the main run (about 3 GB) are included in the repository under `runs/main/`, together with a small local browser (`./browse.sh`) that shows, for any task and condition, the transcript, the evaluator's verdict with its state diff, and the tool schemas exactly as the model saw them, the 200 files of the repeated Claude Haiku 5.5 baseline run under `runs/main/rerun/`, the 3 600 files of the reasoning-controlled pairs under `runs/main/reasoning/`, and the 2 184 ToolSandbox plays (each with ToolSandbox's own conversation record and execution context next to our trajectory file) under `runs/main/toolsandbox/`; the browser covers both benchmarks and the reasoning variants.

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This study used the Berkeley Function Calling Leaderboard harness, backends and evaluator unchanged (Gorilla LLM contributors, 2026; Patil et al., 2025); we thank its authors for releasing them, and likewise the authors of ToolSandbox (Lu et al., 2024), used unchanged. Models were accessed through OpenRouter (OpenRouter, 2026).

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A Additional tables

Table 11: Documentation sensitivity score $S_m = \frac{1}{N}\sum_i (P_{m,0} - P_{m,i})/P_{m,0}$, the mean relative drop from D0 over the single ablations D1–D4 and over D1–D5. Lower is more robust. The score averages a large D2 loss with near-zero or negative changes elsewhere, so it understates every model's dependence on parameter descriptions; it is reported for completeness only.

Model	S (D1–D4)	S (D1–D5)
GPT-6 Luna	0.103	0.279
Claude Haiku 5.5	0.002	0.202
Qwen3-30B-A3B	0.209	0.367
Gemma 4 26B-A4B	0.118	0.294

Table 12: ToolSandbox: every ablation against D0, paired over the 91 scenarios. Similarity columns are means (×100) with the bootstrap 95% CI of the paired difference and a sign-flip permutation p (10 000 permutations); the exact-match columns give the share of scenarios at similarity 1.0, the discordant cells (solved exactly only in D0 / only in the ablation) and the exact McNemar p.

		mean similarity					solved exactly			
Model	Cond.	D0	abl.	Δ	95% CI	p	D0 %	abl. %	D0/abl. only	McNemar p
GPT-6 Luna	D1	81.5	82.9	+1.4	[-4.5, 7.4]	0.6518	25.3	31.9	5/11	0.2101
	D2	81.5	81.2	-0.3	[-5.7, 5.6]	0.9139	25.3	25.3	5/5	1.0000
	D3	81.5	85.4	+3.9	[-1.9, 9.9]	0.2012	25.3	31.9	5/11	0.2101
	D4	81.5	82.5	+1.0	[-5.0, 7.3]	0.7450	25.3	29.7	5/9	0.4240
	D5	81.5	32.3	-49.2	[-59.0, -39.4]	<0.0001	25.3	14.3	14/4	0.0309
Claude Haiku 5.5	D1	80.1	82.1	+2.0	[-3.4, 7.6]	0.4983	26.4	29.7	1/4	0.3750
	D2	80.1	82.2	+2.0	[-0.8, 5.4]	0.2035	26.4	25.3	2/1	1.0000
	D3	80.1	79.3	-0.8	[-5.0, 3.6]	0.7176	26.4	25.3	2/1	1.0000
	D4	80.1	81.8	+1.6	[-2.2, 6.5]	0.4793	26.4	29.7	1/4	0.3750
	D5	80.1	36.3	-43.8	[-54.8, -32.2]	<0.0001	26.4	17.6	16/8	0.1516
Qwen3-30B-A3B	D1	65.0	60.8	-4.2	[-13.3, 4.4]	0.3525	8.8	14.3	3/8	0.2266
	D2	65.0	58.7	-6.3	[-13.1, 0.4]	0.0706	8.8	9.9	3/4	1.0000
	D3	65.0	62.4	-2.5	[-9.6, 3.9]	0.4796	8.8	11.0	3/5	0.7266
	D4	65.0	69.0	+4.1	[-3.3, 11.7]	0.3038	8.8	14.3	3/8	0.2266
	D5	65.0	26.3	-38.7	[-51.1, -26.2]	<0.0001	8.8	14.3	5/10	0.3018
Gemma 4 26B-A4B	D1	75.5	73.7	-1.8	[-5.1, 1.3]	0.2700	19.8	13.2	6/0	0.0312
	D2	75.5	76.5	+0.9	[-2.9, 4.6]	0.6534	19.8	19.8	2/2	1.0000
	D3	75.5	69.5	-6.0	[-10.4, -2.3]	0.0029	19.8	12.1	7/0	0.0156
	D4	75.5	75.0	-0.5	[-5.8, 4.2]	0.8380	19.8	17.6	5/3	0.7266
	D5	75.5	27.3	-48.2	[-61.2, -34.5]	<0.0001	19.8	16.5	15/12	0.7011

Table 13: ToolSandbox mean similarity (×100) by scenario category, D0 against D2. Categories are ToolSandbox's own and overlap (n is the number of the 91 scenarios in each).

		GPT-6 Luna		Claude Haiku 5.5		Qwen3-30B-A3B		Gemma 4 26B-A4B	
Category	n	D0	D2	D0	D2	D0	D2	D0	D2
single tool call	13	90.2	89.8	92.8	92.7	84.6	79.9	91.3	91.4
multiple tool calls	60	81.9	81.4	86.3	89.3	72.3	64.5	87.6	89.0
single user turn	53	84.0	84.8	88.1	90.6	75.3	67.5	88.4	91.1
multiple user turns	20	81.6	77.8	85.5	88.1	72.3	66.7	87.9	85.0
state dependency	13	74.5	73.3	78.7	86.4	80.3	74.7	87.2	88.7
canonicalization	36	75.8	76.8	88.6	91.8	70.0	58.5	87.4	90.0
insufficient information	18	74.1	74.3	50.7	50.9	26.2	23.9	23.9	24.0

Table 14: Earliest divergence among failed tasks (counts). "W.A.": first unmatched ground-truth call was made with wrong arguments; "N.C.": the tool was never called; "None": every ground-truth call was matched but the task still failed.

	D0			D2			D5		
Model	W.A.	N.C.	None	W.A.	N.C.	None	W.A.	N.C.	None
GPT-6 Luna	30	56	2	69	65	1	56	138	4
Claude Haiku 5.5	36	47	16	70	52	5	10	188	2
Qwen3-30B-A3B	44	43	4	95	45	3	78	120	2
Gemma 4 26B-A4B	33	20	3	88	31	3	33	167	0

Table 15: Recovery after an execution error: share of tasks containing at least one failed call that still passed, with the number of such tasks in parentheses.

Model	D0	D1	D2	D3	D4	D5
GPT-6 Luna	100% (1)	67% (3)	12% (24)	– (0)	33% (3)	0% (38)
Claude Haiku 5.5	50% (2)	– (0)	22% (23)	33% (3)	100% (1)	0% (17)
Qwen3-30B-A3B	25% (4)	0% (4)	20% (30)	8% (12)	0% (5)	0% (102)
Gemma 4 26B-A4B	33% (3)	– (0)	21% (29)	50% (4)	0% (1)	0% (30)

Table 16: Tool-use behaviour, mean per task over all 200 tasks (failures included). Calls are BFCL-decoded tool calls; steps are model requests; a repeated call has identical tool and arguments to an earlier call in the same task; "calls/pass" is the mean number of calls over passing tasks only.

Model	Cond.	calls	steps	on-path	off-path	failed	repeated	distinct tools	calls/pass
GPT-6 Luna	D0	7.4	10.2	5.8	1.6	0.01	0.50	6.2	7.4
	D1	7.5	10.2	5.6	1.9	0.01	0.58	6.0	7.2
	D2	7.6	10.6	5.6	1.9	0.15	0.52	6.0	7.8
	D3	7.6	10.4	5.9	1.7	0.00	0.65	6.2	7.6
	D4	7.7	10.5	5.8	1.8	0.01	0.59	6.3	7.5
	D5	12.9	14.7	4.2	8.4	0.28	0.86	9.1	22.5
Claude Haiku 5.5	D0	7.9	10.1	6.1	1.8	0.01	0.70	6.3	7.6
	D1	7.3	9.7	5.7	1.6	0.00	0.52	6.0	6.9
	D2	8.4	10.6	6.4	1.8	0.12	0.85	6.2	7.8
	D3	7.6	10.0	6.0	1.7	0.01	0.58	6.2	7.4
	D4	8.0	10.2	6.2	1.9	0.01	0.78	6.4	8.0
	D5	4.7	7.3	1.4	3.2	0.10	0.25	3.5	–
Qwen3-30B-A3B	D0	8.0	11.7	6.2	1.9	0.02	0.85	6.5	7.4
	D1	7.8	11.3	5.9	1.8	0.07	0.81	6.3	6.8
	D2	9.4	12.9	6.5	2.4	0.51	1.38	6.6	7.1
	D3	7.9	11.4	6.0	1.8	0.07	0.86	6.4	7.1
	D4	8.3	11.9	6.2	2.1	0.04	1.14	6.4	6.4
	D5	19.5	21.1	4.0	14.4	1.09	3.24	15.6	–
Gemma 4 26B-A4B	D0	8.3	11.4	6.6	1.8	0.02	1.24	6.3	7.4
	D1	8.3	11.5	6.7	1.7	0.00	1.29	6.2	7.4
	D2	11.1	13.8	8.5	2.4	0.23	3.38	5.9	7.2
	D3	8.3	11.4	6.6	1.6	0.02	1.27	6.3	7.4
	D4	8.5	11.6	6.8	1.8	0.01	1.32	6.2	7.8
	D5	11.0	14.0	3.9	6.4	0.73	3.94	5.0	–

Table 17: Terminal error assigned by BFCL per failed task. "State": wrong final state; "Output": required output missing; "Empty": turn ended without a tool call; "Limit": 20-step limit reached; "Provider": inference error from the provider.

Model	Cond.	State	Output	Empty	Limit	Provider	Total failed
GPT-6 Luna	D0	61	17	10	0	0	88
	D1	63	18	14	0	0	95
	D2	96	30	9	0	0	135
	D3	60	15	10	0	0	85
	D4	61	13	9	0	0	83
	D5	93	34	50	21	0	198
Claude Haiku 5.5	D0	64	16	19	0	0	99
	D1	63	15	26	0	0	104
	D2	86	24	17	0	0	127
	D3	57	16	16	0	0	89
	D4	49	9	19	0	0	77
	D5	78	43	78	1	0	200
Qwen3-30B-A3B	D0	64	21	2	3	1	91
	D1	69	27	5	2	0	103
	D2	97	34	6	6	0	143
	D3	64	24	2	4	1	95
	D4	94	14	3	3	0	114
	D5	103	33	5	57	2	200
Gemma 4 26B-A4B	D0	35	15	2	4	0	56
	D1	34	20	4	4	0	62
	D2	76	26	0	20	0	122
	D3	34	17	1	5	0	57
	D4	34	14	0	3	0	51
	D5	94	54	17	35	0	200

B Case-study trajectories

The four case studies of Section 4.5, selected by the rule given there, with every decoded tool call and its execution result under D0, D2 and D5. Results are truncated to about 100 characters; execution errors are shown in red. The final reply is the model's last plain-text message to the user.

GPT-6 Luna, task `multi_turn_base_177`

Turn 0: Consider this: A patient, a traveler in this context, wants to book a flight using the card with id 'card_7629' and the access token 'abc123xyz', expiring on 10/2026, with CVV 456. They intend to fly from Los Angeles to Tokyo on December 15, 2026, in business class, estimated at $1,200.

Turn 1: Reflect on the importance of verifying information. Our traveler needs to confirm their flight booking's details for assurance.

Ground truth. t0: `book_flight(access_token='abc123xyz', card_id='card_7629', travel_date='2026-12-15', travel_from='LAX', travel_to='HND', travel_class='business')`; t1: `retrieve_invoice(access_token='abc123xyz', booking_id='3426812')`

	D0 original **pass** (6 steps, 5 calls)	D2 no parameter descriptions **fail**, required output missing (6 steps, 6 calls)	D5 all removed **fail**, turn ended without a call (2 steps, 0 calls)
t0	`get_nearest_airport_by_city({"location": "Los Angeles"})` `-> {"nearest_airport": "LAX"}`	`get_nearest_airport_by_city({"location": "Los Angeles"})` `-> {"nearest_airport": "LAX"}`	*(no tool call)*
	`get_nearest_airport_by_city({"location": "Tokyo"})` `-> {"nearest_airport": "HND"}`	`get_nearest_airport_by_city({"location": "Tokyo"})` `-> {"nearest_airport": "HND"}`	
	`get_flight_cost({"travel_from": "LAX", "travel_to": "HND", "travel_date": "2026-12-15", "travel_class": "business"})` `-> {"travel_cost_list": [860.0]}`	`get_flight_cost({"travel_from": "Los Angeles", "travel_to": "Tokyo", "travel_date": "2026-12-15", "travel_class": "business"})` `-> Error during execution: No available route for the given airports.`	
	`book_flight({"access_token": "abc123xyz", "card_id": "card_7629", "travel_date": "2026-12-15", "travel_from": "LAX", "travel_to": "HND", "travel_class": "busin...)` `-> {"booking_id": "3426812", "transaction_id": "45451592", "booking_status": true, "booking_history"...`	`get_flight_cost({"travel_from": "LAX", "travel_to": "HND", "travel_date": "2026-12-15", "travel_class": "business"})` `-> {"travel_cost_list": [860.0]}`	

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	D0 original **pass** (6 steps, 5 calls)	D2 no parameter descriptions **fail**, required output missing (6 steps, 6 calls)	D5 all removed **fail**, turn ended without a call (2 steps, 0 calls)
		`book_flight({"access_token": "abc123xyz", "card_id": "card_7629", "travel_date": "2026-12-15", "travel_from": "LAX", "travel_to": "HND", "travel_class": "busin...)` `-> {"booking_id": "3426812", "transaction_id": "45451592", "booking_status": true, "booking_history"...`	
t1	`retrieve_invoice({"access_token": "abc123xyz", "booking_id": "3426812"})` `-> {"invoice": {"booking_id": "3426812", "travel_date": "2026-12-15", "travel_from": "LAX", "travel_...`	`get_booking_history({"access_token": "abc123xyz"})` `-> {"booking_history": {"3426812": {"card_id": "card_7629", "travel_date": "2026-12-15", "travel_fro...`	*(no tool call)*
	Final reply: "I verified the booking invoice. It confirms: - **Route:** Los Angeles (LAX) to Tokyo Haneda (HND) - **Date:** December 15, 2026 - **Class:** Business - **Fare:** $860 - **Booking ID:** 3426812 - **Transaction ID:** 45451592"	*Final reply:* "I verified the booking against the travel system. Booking **3426812** is listed as: - **Route:** Los Angeles (LAX) → Tokyo Haneda (HND) - **Date:** December 15, 2026 - **Class:** Business - **Recorded fare:** $860 - **Card:** card_7629 - **Transaction ID:** 45451592 These details match the booking..."	*Final reply:* "Verifying the itinerary before relying on it helps prevent costly mistakes—especially with an international trip. Check the exact travel date, departure and arrival airports, times and time zones, business-class cabin, passenger name, baggage rules, and change or cancellation terms. Confirm that the..."

Claude Haiku 5.5, task `multi_turn_base_78`

Turn 0: I'm gearing up for a quick business getaway and need my ride all set. Would you be able to verify if my tire pressure is in check? If it falls under 37.5 PSI, perhaps we could swing by the nearest tire shop?

Turn 1: While we head over to ensure my tires are roadworthy, I'd like to send out a swift update on my business account. Let's post a tweet: 'Ensuring my wheels are well-maintained. Maintenance is key to success!' with the hashtag 'BusinessOnTheMove'.

Ground truth. t0: `check_tire_pressure()`, `find_nearest_tire_shop()`; t1: `post_tweet(content='Ensuring my wheels are well-maintained. Maintenance is key to success!', tags=['#BusinessOnTheMove'])`

	D0 original **pass** (5 steps, 3 calls)	D2 no parameter descriptions **fail**, wrong final state (5 steps, 3 calls)	D5 all removed **fail**, turn ended without a call (2 steps, 0 calls)

t0	`check_tire_pressure({})`
	`-> {"frontLeftTirePressure": 35.0, "frontRightTirePressure": 35.0, "rearLeftTirePressure": 33.0, "re...`
	`find_nearest_tire_shop({})`
	`-> {"shopLocation": "456 Oakwood Avenue, Rivermist, 83214"}`
		`check_tire_pressure({})`
		`-> {"frontLeftTirePressure": 35.0, "frontRightTirePressure": 35.0, "rearLeftTirePressure": 33.0, "re...`
		`find_nearest_tire_shop({})`
		`-> {"shopLocation": "456 Oakwood Avenue, Rivermist, 83214"}`
			(no tool call)

t1	`post_tweet({"content": "Ensuring my wheels are well-maintained. Maintenance is key to success!", "tags": ["#BusinessOnTheMove"]})` `-> {"id": 10, "username": "businesspro", "content": "Ensuring my wheels are well-maintained. Mainten...`	`post_tweet({"content": "Ensuring my wheels are well-maintained. Maintenance is key to success! #BusinessOnTheMove"})` `-> {"id": 10, "username": "businesspro", "content": "Ensuring my wheels are well-maintained. Mainten...`	*(no tool call)*
	Final reply: "Your tweet is posted on the account @businesspro: "Ensuring my wheels are well-maintained. Maintenance is key to success!" #BusinessOnTheMove I haven't set navigation to the tire shop yet. Do you want me to send you to 456 Oakwood Avenue, Rivermist, 83214?"	*Final reply:* "Your tweet is posted from the businesspro account: "Ensuring my wheels are well-maintained. Maintenance is key to success! #BusinessOnTheMove" Do you want me to set navigation to the tire shop at 456 Oakwood Avenue, Rivermist, 83214?"	*Final reply:* "I can't post that yet. None of my tools are labeled as social media or tweeting functions, so I'd be guessing which one to use, and posting is a public action I don't want to get wrong. Your tweet is ready to go: "Ensuring my wheels are well-maintained. Maintenance is key to success! #BusinessOnTh..."

Qwen3-30B-A3B, task `multi_turn_base_42`

Turn 0: In my existing folder called 'Lectures', I'd like to set up a new file called 'Notes2023.txt'.

Turn 1: Inside 'Notes2023.txt', I shall jot down some pivotal reminders: 'Study diligently, practice programming, master algorithms.'

Turn 2: Ascertain for me how many characters of text remain in this alphabetically sorted version of 'Notes2023.txt'.

Ground truth. t0: `touch(file_name='Notes2023.txt')`; t1: `echo(content='Study diligently, practice programming, master algorithms.', file_name='Notes2023.txt')`; t2: `wc(file_name='Notes2023.txt', mode='c')`

	D0 original **pass** (11 steps, 8 calls)	D2 no parameter descriptions **fail**, wrong final state (12 steps, 9 calls)	D5 all removed **fail**, wrong final state (14 steps, 11 calls)
t0	`cd({"folder": "Lectures"})` `-> {"error": "cd: 'Lectures': No such file or directory"}`	`cd({"folder": "Lectures"})` `-> {"error": "cd: 'Lectures': No such file or directory"}`	`cat({"file_name": "Lectures/Notes2023.txt"})` `-> {"error": "cat: 'Lectures/Notes2023.txt': Invalid character"}`
	`ls({})` `-> {"current_directory_content": []}`	`ls({})` `-> {"current_directory_content": []}`	`cd({"folder": "Lectures/Notes2023.txt"})` `-> {"error": "cd: Lectures/Notes2023.txt: Unsupported path. Only one folder level at a time is suppo...`
	`pwd({})` `-> {"current_working_directory": "/Lectures"}`		`cp({"source": "Lectures", "destination": "exists"})` `-> {"error": "cp: cannot copy 'Lectures': No such file or directory"}`
	`ls({})` `-> {"current_directory_content": []}`		`diff({"file_name1": "Lectures", "file_name2": "create"})` `-> {"error": "diff: Lectures or create: No such file or directory"}`
	`touch({"file_name": "Notes2023.txt"})` `-> None`		`du({})` `-> {"disk_usage": "0 bytes"}`
			`echo({"content": "Lectures"})` `-> {"terminal_output": "Lectures"}`
			`find({"path": "Lectures/Notes2023.txt"})` `-> {"error": "find: 'Lectures/Notes2023.txt': No such file or directory"}`
			`grep({"file_name": "Lectures", "pattern": "list"})` `-> {"error": "grep: Lectures: No such file or directory"}`
			`ls({})` `-> {"current_directory_content": []}`
			`mkdir({"dir_name": "Lectures"})` `-> None`
			`mv({"source": "Notes2023.txt", "destination": "create"})` `-> {"error": "mv: cannot move 'Notes2023.txt': No such file or directory"}`

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	D0 original **pass** (11 steps, 8 calls)	D2 no parameter descriptions **fail**, wrong final state (12 steps, 9 calls)	D5 all removed **fail**, wrong final state (14 steps, 11 calls)

t1	`echo({"content": "Study diligently, practice programming, master algorithms.", "file_name": "Notes2023.txt"})`
	`-> None`
		`mkdir({"dir_name": "Lectures"})`
		`-> None`
		`cd({"folder": "Lectures"})`
		`-> {"current_working_directory": "Lectures"}`
		`echo({"content": "Study diligently, practice programming, master algorithms.", "file_name": "Notes2023.txt"})`
		`-> {"error": "echo: cannot write to 'Notes2023.txt': No such file"}`
		`touch({"file_name": "Notes2023.txt"})`
		`-> None`
		`echo({"content": "Study diligently, practice programming, master algorithms.", "file_name": "Notes2023.txt"})`
		`-> None`
			(no tool call)

t2	`sort({"file_name": "Notes2023.txt"})`
	`-> {"sorted_content": "Study diligently, practice programming, master algorithms."}`
	`wc({"file_name": "Notes2023.txt", "mode": "c"})`
	`-> {"count": 58, "type": "characters"}`
		`sort({"file_name": "Notes2023.txt"})`
		`-> {"sorted_content": "Study diligently, practice programming, master algorithms."}`
		`wc({"file_name": "Notes2023.txt", "mode": "c"})`
		`-> {"count": 58, "type": "characters"}`
			(no tool call)

	Final reply: "The alphabetically sorted version of 'Notes2023.txt' contains 58 characters. This includes all letters, spaces, and punctuation marks in the text "Study diligently, practice programming, master algorithms.""	*Final reply:* "I've analyzed the 'Notes2023.txt' file and found the following: The file contains 58 characters in total (including spaces and punctuation). When sorted alphabetically, the content remains the same: "Study diligently, practice programming, master algorithms." This is because the text is already in..."	*Final reply:* "It seems that the file 'Notes2023.txt' could not be created or accessed due to system limitations. As a result, there is no content to analyze or count characters from. If you'd like, I can help you draft the text you want to include in the file, or guide you through creating it in a different envi..."

Gemma 4 26B-A4B, task `multi_turn_base_11`

Turn 0: Display all the available files located within the '/temp' directory including hidden one using the terminal interface .

Turn 1: With the information presented on the terminal, draft a post on Twitter mentioning only the

file names seperated them by comma and space, and include the hashtag '#fileshowcase' to spread the word.

Ground truth. t0: `ls(a=True)`; t1: `post_tweet(content='file1.txt, file2.txt', tags=['#fileshowcase'])`

	D0 original **pass** (15 steps, 13 calls)	D2 no parameter descriptions **fail**, wrong final state (13 steps, 11 calls)	D5 all removed **fail**, required output missing (3 steps, 1 calls)
t0	`cd({"folder": "temp"})` `-> {"error": "cd: 'temp': No such file or directory"}`	`ls({"a": true})` `-> {"current_directory_content": ["file1.txt", "file2.txt"]}`	`get_user_stats({"username": "ls -a /temp"})` `-> {"tweet_count": 0, "following_count": 0, "retweet_count": 0}`
	`ls({"a": true})` `-> {"current_directory_content": ["file1.txt", "file2.txt"]}`	`cd({"folder": "/temp"})` `-> {"error": "cd: /temp: Unsupported path. Only one folder level at a time is supported."}`	
	`find({"path": "/temp"})` `-> {"error": "find: '/temp': No such file or directory"}`	`ls({"a": true})` `-> {"current_directory_content": ["file1.txt", "file2.txt"]}`	
	`ls({"a": true})` `-> {"current_directory_content": ["file1.txt", "file2.txt"]}`	`find({"path": "/temp"})` `-> {"error": "find: '/temp': No such file or directory"}`	
	`find({"path": "/"})` `-> {"matches": ["/file1.txt", "/file2.txt"]}`	`ls({"a": true})` `-> {"current_directory_content": ["file1.txt", "file2.txt"]}`	
	`ls({"a": true})` `-> {"current_directory_content": ["file1.txt", "file2.txt"]}`	`find({"path": "."})` `-> {"matches": ["./file1.txt", "./file2.txt"]}`	
	`find({"path": "/"})` `-> {"matches": ["/file1.txt", "/file2.txt"]}`	`ls({"a": true})` `-> {"current_directory_content": ["file1.txt", "file2.txt"]}`	
	`find({"path": "/"})` `-> {"matches": ["/file1.txt", "/file2.txt"]}`	`find({"path": "/"})` `-> {"matches": ["/file1.txt", "/file2.txt"]}`	
	`find({"path": "/"})` `-> {"matches": ["/file1.txt", "/file2.txt"]}`	`find({"path": "/"})` `-> {"matches": ["/file1.txt", "/file2.txt"]}`	
	`find({"path": "/"})` `-> {"matches": ["/file1.txt", "/file2.txt"]}`	`find({"path": "/temp"})` `-> {"error": "find: '/temp': No such file or directory"}`	
	`find({"name": "temp", "path": "/"})` `-> {"matches": []}`		
	`ls({"a": true})` `-> {"current_directory_content": ["file1.txt", "file2.txt"]}`		

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	D0 original **pass** (15 steps, 13 calls)	D2 no parameter descriptions **fail**, wrong final state (13 steps, 11 calls)	D5 all removed **fail**, required output missing (3 steps, 1 calls)
t1	`post_tweet({"content": "file1.txt, file2.txt", "tags": ["#fileshowcase"]})` `-> {"id": 1, "username": "michael", "content": "file1.txt, file2.txt", "tags": ["#fileshowcase"], "m...`	`post_tweet({"content": "file1.txt, file2.txt #fileshowcase"})` `-> {"id": 1, "username": "michael", "content": "file1.txt, file2.txt #fileshowcase", "tags": [], "me...`	*(no tool call)*
	Final reply: "I have posted the tweet for you: "file1.txt, file2.txt #fileshowcase"."	*Final reply:* "The tweet has been posted successfully: "file1.txt, file2.txt #fileshowcase""	*Final reply:* "Since no files were found in the '/temp' directory, I cannot list any file names. If you can provide the file names or if they appear after a different command, I will be happy to draft the tweet for you!"